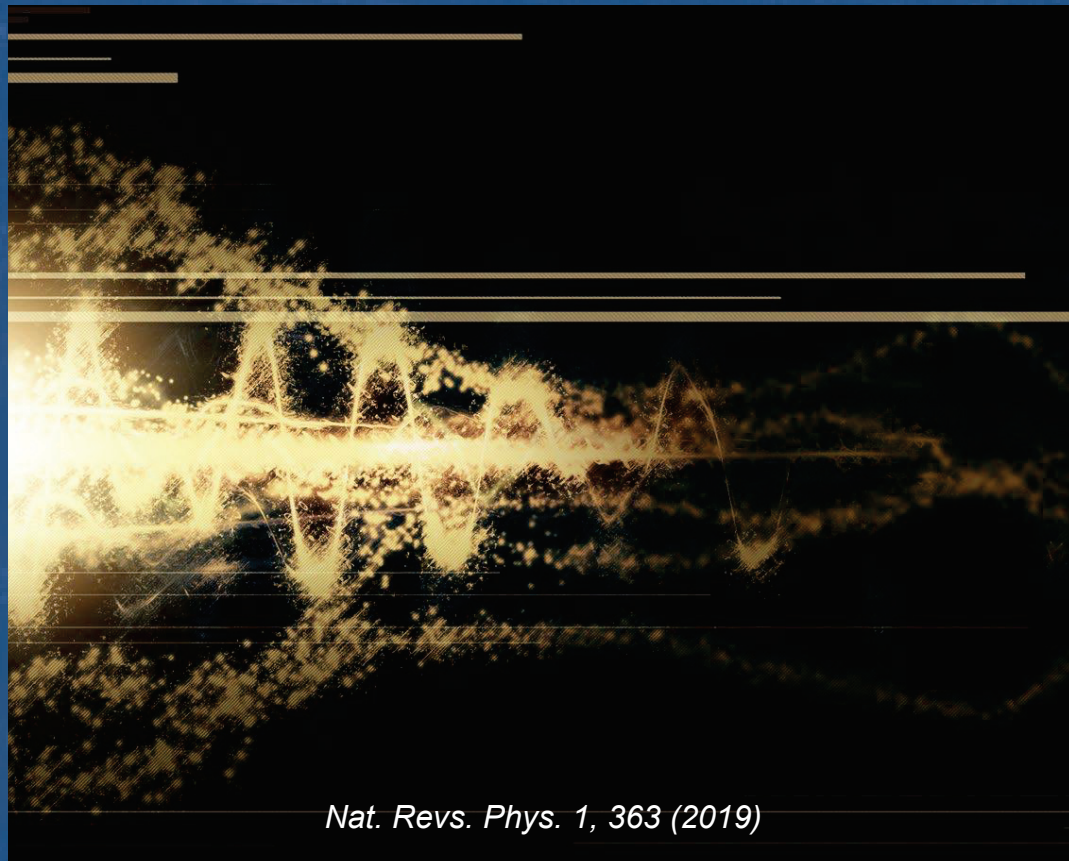


Ultrafast relativistic electron nanoprobe



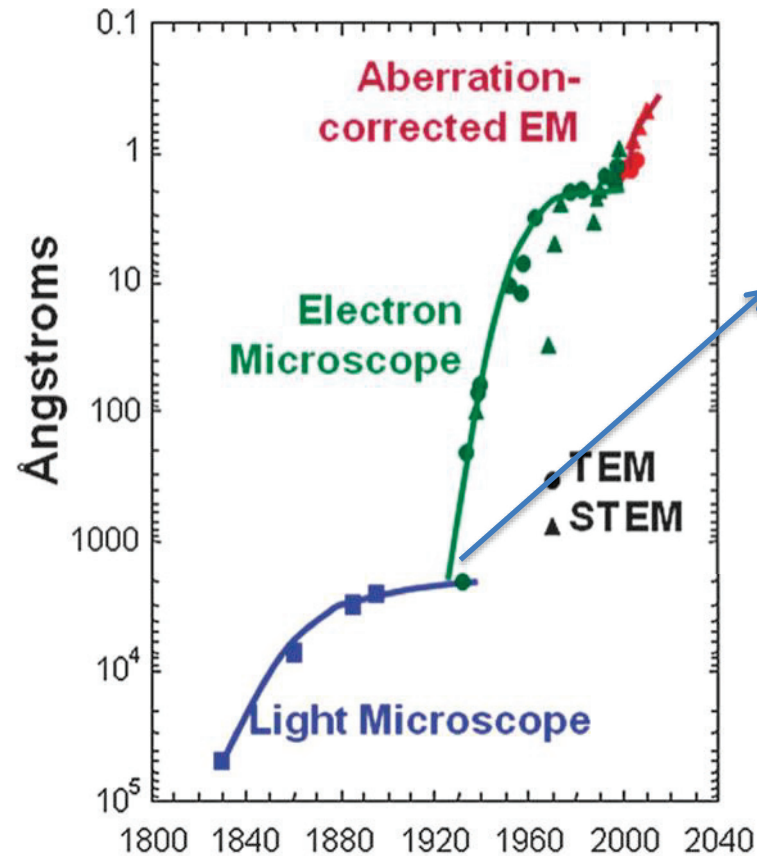
Nat. Revs. Phys. 1, 363 (2019)

Fuhao Ji
NAPAC 2019
Sept 3rd, 2019

Outline

- Introduction
- Challenges in electron-based instrumentation
 - Limitations of current relativistic electron probes
- HiRES
 - The APEX gun
 - HiRES beamline
 - Experiments at HiRES
- Ultrafast relativistic electron nanoprobe
 - Lateral squeezing of ultrafast electron beams
 - Characterization of electron beam near the waist
 - Ultrafast point projection microscopy
 - Ultrafast scanning electron diffraction
- Discussion and Further developments

The electron microscope revolution: Seeing the ultra small



The Nobel Prize in Physics
1937



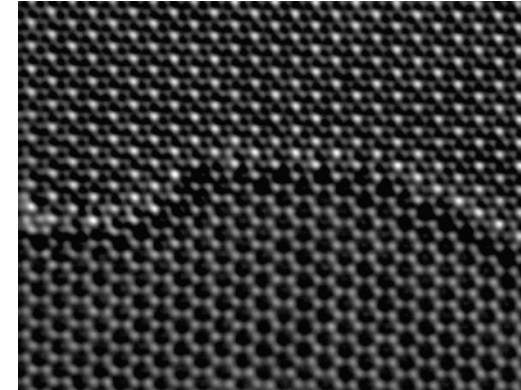
Clinton Joseph
Davisson
Prize share: 1/2



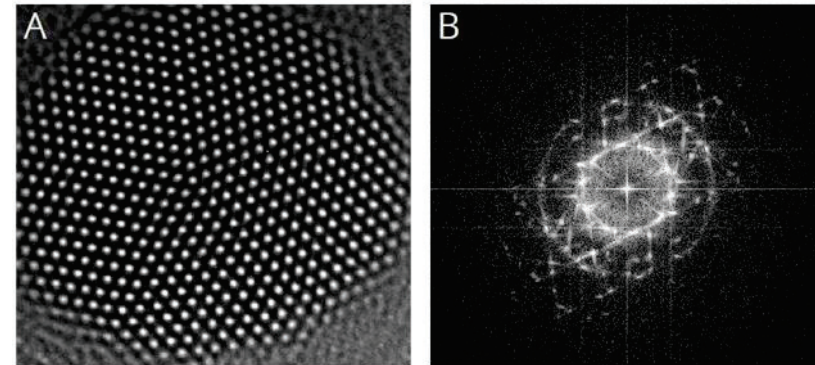
George Paget
Thomson
Prize share: 1/2

$$\lambda = h / p$$

Single/bilayer
graphene



Single gold nanoparticle



Discovered by accident. Davisson & Germer Phys. Rev. 30, 705 (1927)

Fundamental time scales in nature

Atomic Structural Dynamics

atomic vibrational period:

$$T_{\text{vib}} = 2\pi(k/m)^{-1/2} \sim 100 \text{ fs}$$

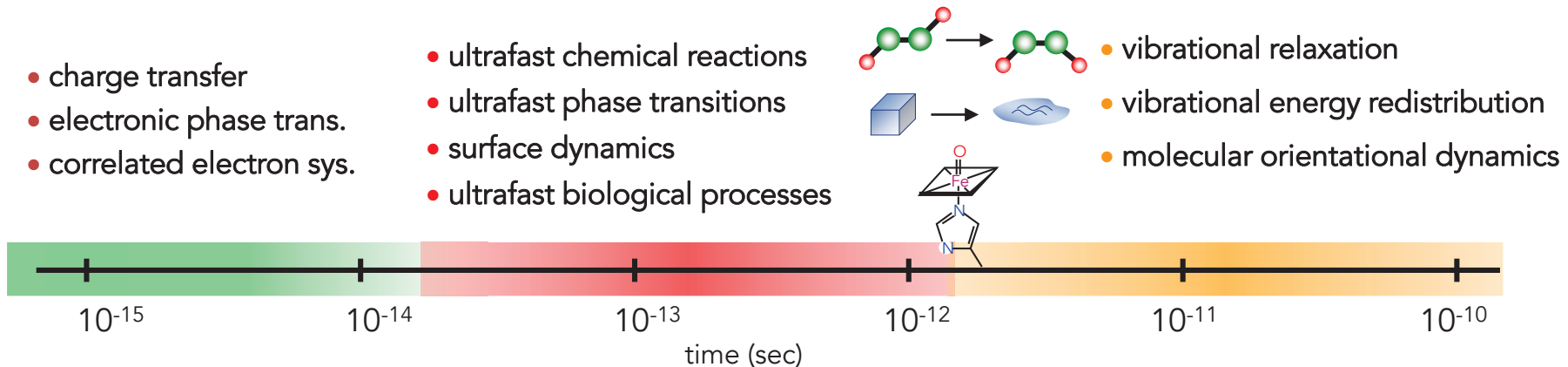
$$k \sim \text{eV}/\text{\AA}^2 \quad m \sim 10^{-25} \text{ Kg}$$

Electronic Structural Dynamics

electron-phonon interaction $\sim 1 \text{ ps}$

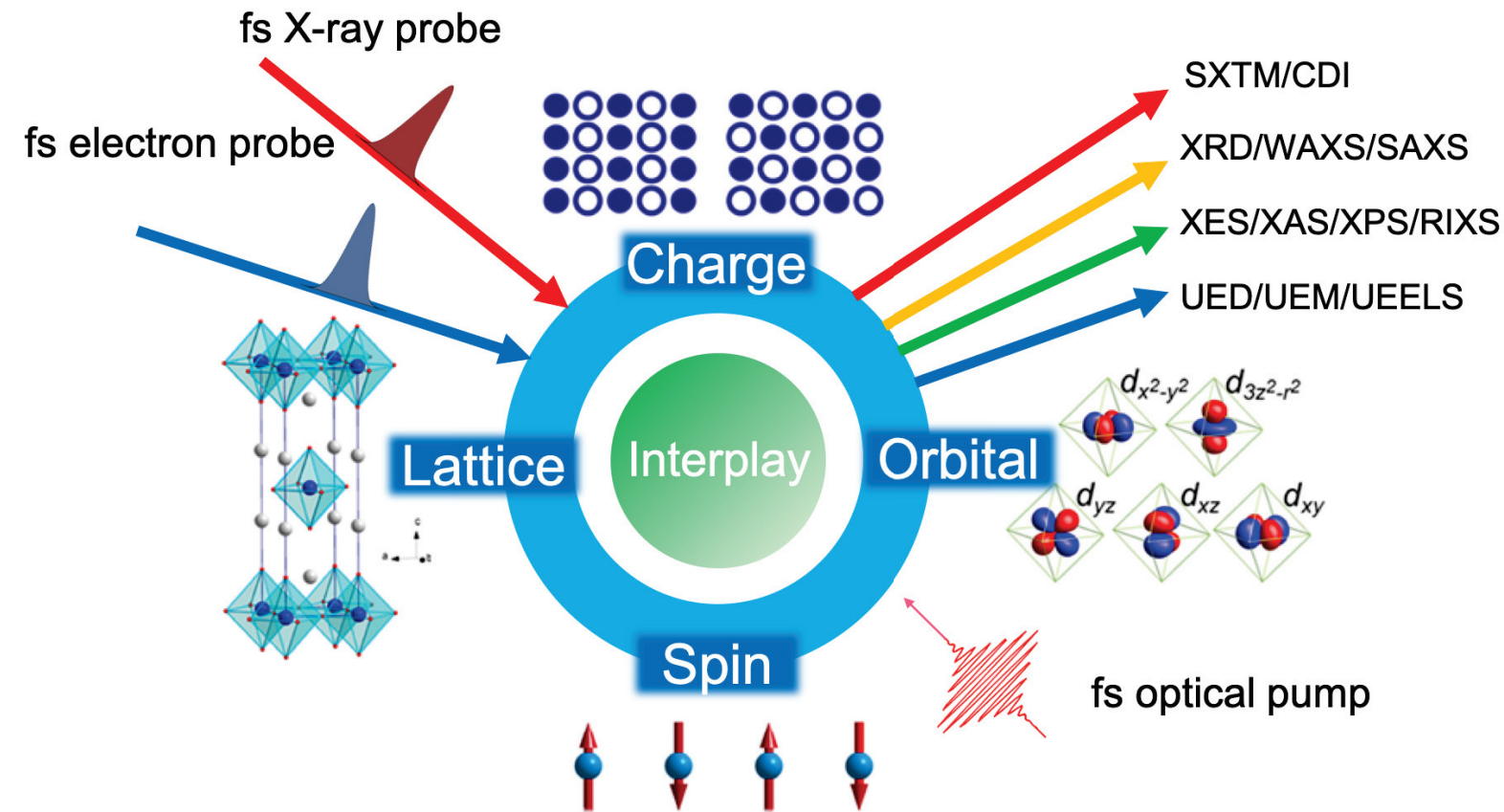
e-e scattering $\sim 10 \text{ fs}$

e⁻ correlation time $\sim 100 \text{ atto } (a/V_{\text{Fermi}})$

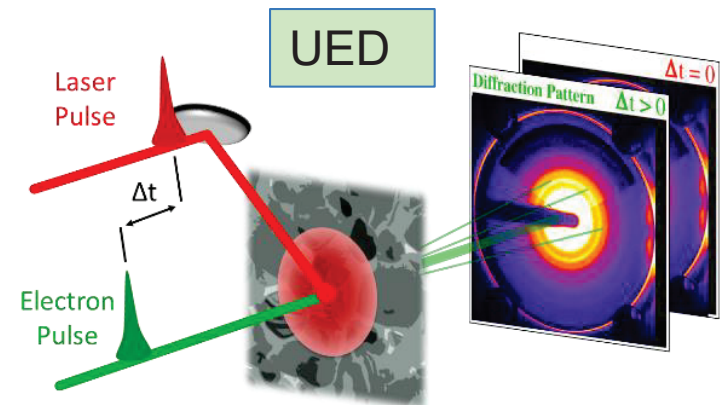


- real-time observation of reaction pathways
- separate correlated phenomena in the time domain
- direct observations of the underlying correlations as they develop

Challenge: understand quantum materials in the atomic time scale



- Strongly entangled subsystems
- Exotic material properties
- Call for ultrafast probes
- Complementary tools: FEL and UES

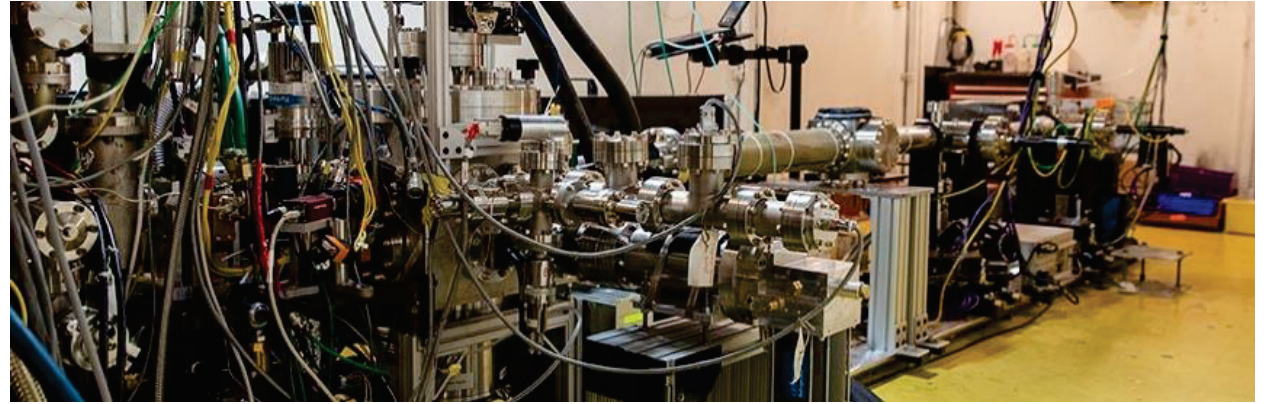


$$\sigma_{el} = \frac{h^2 Z^{4/3} c^2}{\pi E_0^2 \beta^2} = 8.57 \cdot 10^{-20} \text{ cm}^2 \text{ for } C$$

Ultrafast relativistic electron diffraction

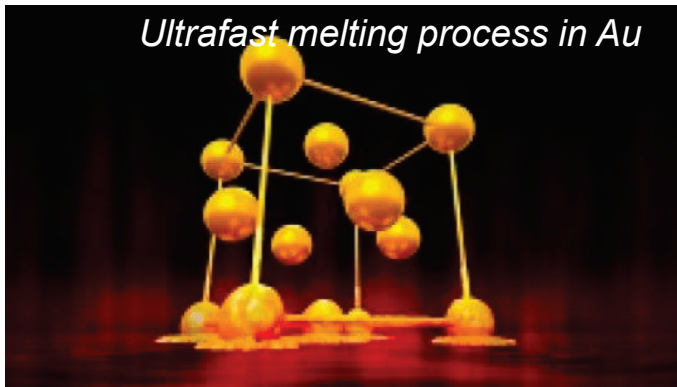
Advantages

- Radio frequency (RF) based electron sources, relativistic beam energy
- Larger penetration depth, reduce multiple scattering
- Shorter wavelength, larger signal in high order peaks
- Maximizing the highest electron density
- Minimizing space charge
- TOA jitter dumped by the relativistic speed
- Minimizing velocity mismatch

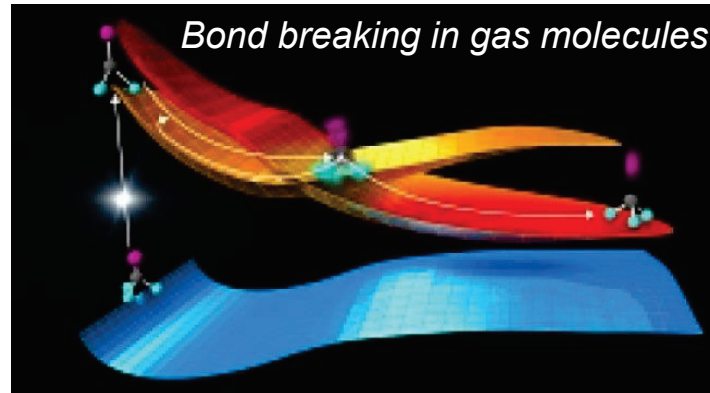


RSI 86, 073702 (2015)

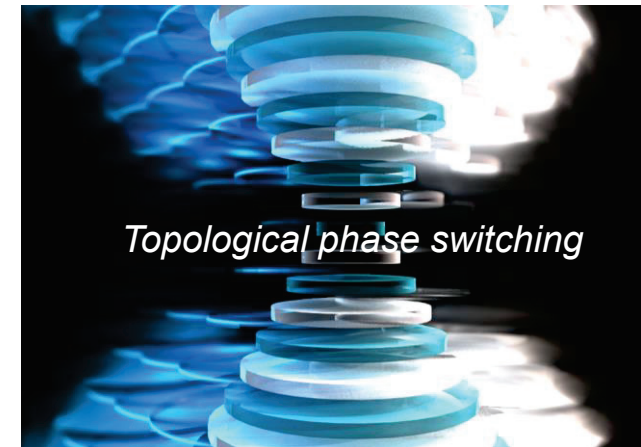
Enormous Scientific results produced



Science 360, 1451 (2018)

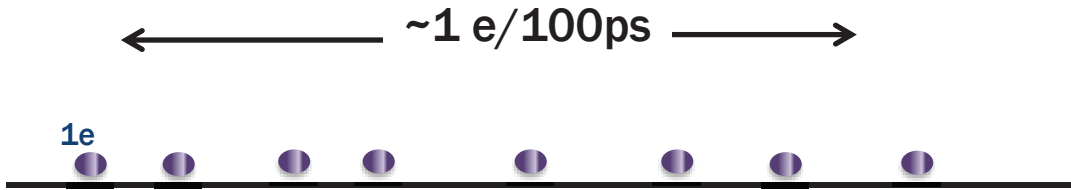


Science 361, 64 (2018)



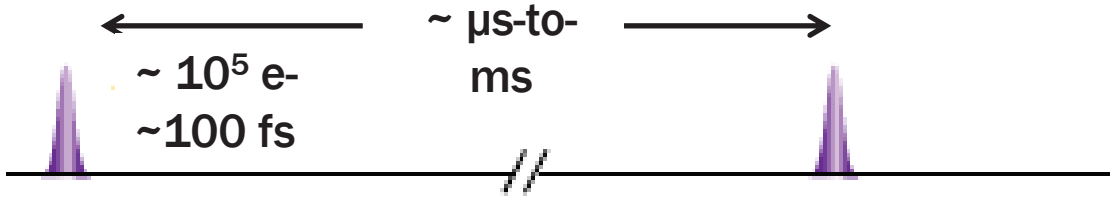
Nature 565, 61 (2019)

Limitation of current relativistic electron probes



High current density
Ultrasmall emittance
High brightness

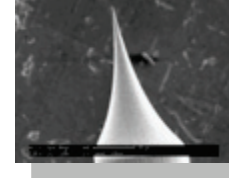
- Sub-nm probes
- meV Energy resolution
- Atomic resolution images



High pulse Charge
Femtosecond pulses

- um-scale probes
- 10-100 eV Energy resolution
- Lower Brightness, larger emittance

TEM source

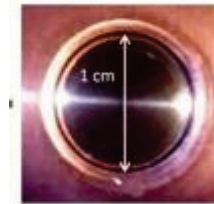


- nm-sized cathodes
- Continuous emission

$$B_{av} \propto \frac{I}{\varepsilon_n^2} = \frac{Q_{pulse} f}{\varepsilon_n^2}$$

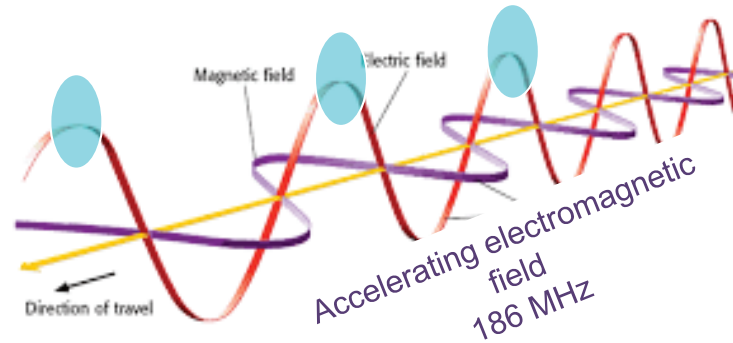
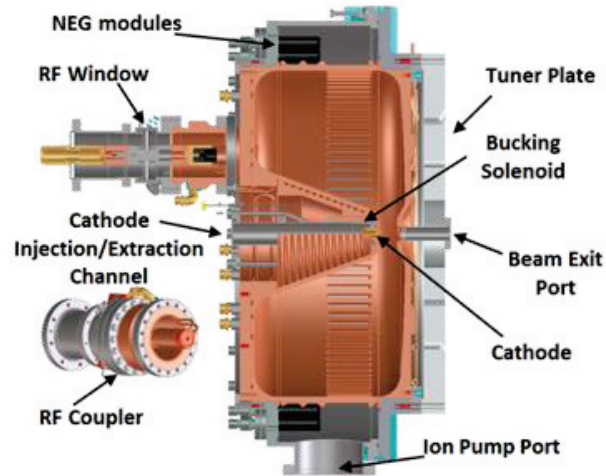
- Tip based DC source take advantage of extremely small source area
- Relativistic sources are limited by low average transverse brightness due to the big emission area
- For pulsed sources, the most straight forward way is to increase repetition rate

Relativistic UED/UEM setups



- mm-sized cathodes
- Pulsed emission

The Advanced Photoinjector Experiment (APEX) project at LBNL



Frequency	186 MHz
Operation mode	CW
Gap voltage	Up to 800 kV
Field at the cathode	> 20 MV/m
Peak wall power density	25.0 W/cm ²
base pressure	~ 3*10 ⁻¹¹ Torr

Nucl. Instr. & Meth. A 599, 9 (2009)

Phys. Rev. ST Accel. Beams 15, 103501 (2012)

Phys. Rev. ST Accel. Beams 18, 013401 (2015)

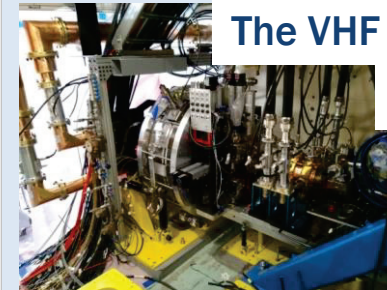
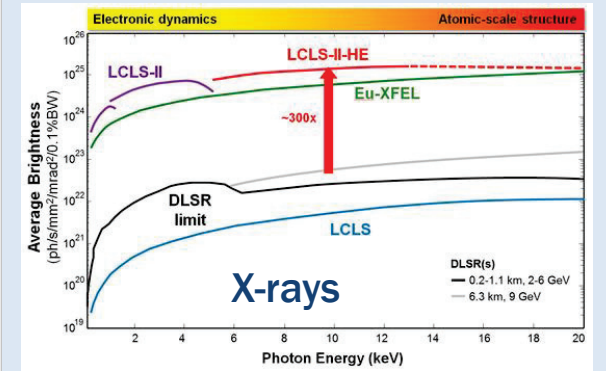
Appl. Phys. Lett. 107, 042104 (2015)

Transverse/ longitudinal space charge:

$$\propto \frac{1}{\gamma^3}, \frac{1}{\gamma^5}$$

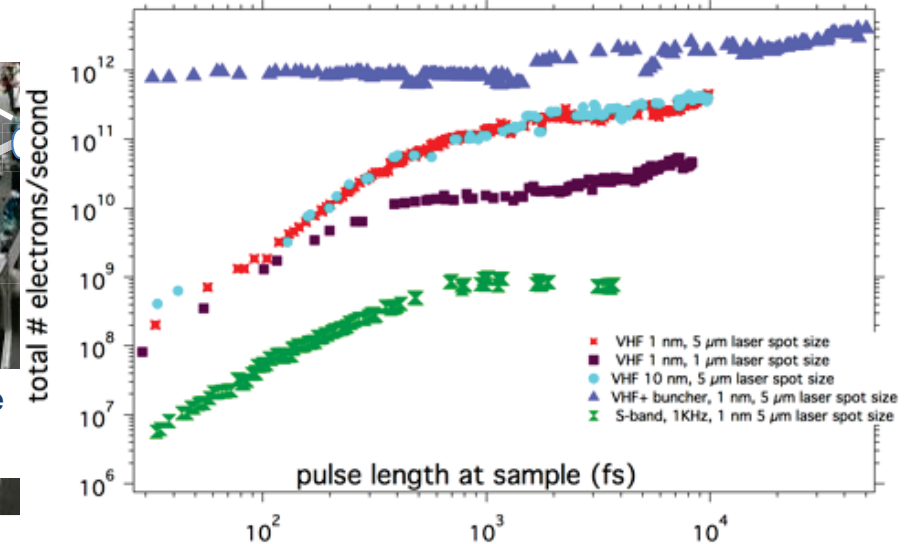
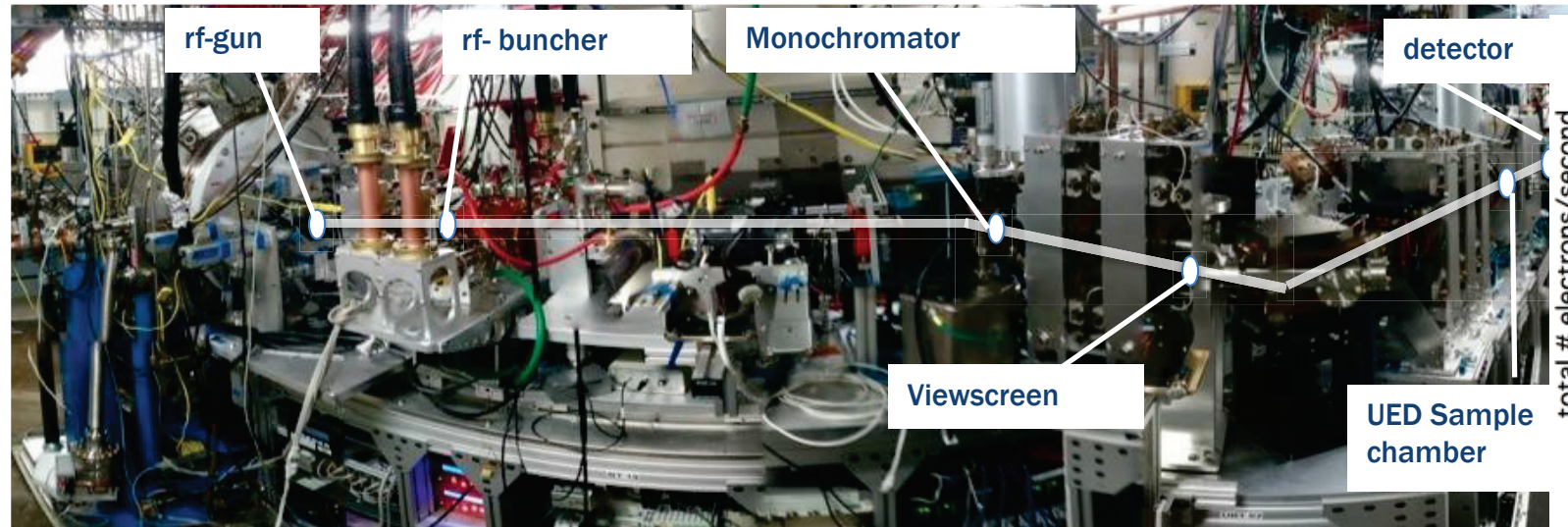
- High fields & ~MeV beam → space charge, brightness, Vel. mismatch
- MHz repetition rate → High flux, low noise
- Normal conducting → Reliable, stable operations
- long wavelength → Large temporal acceptance, from fs to ns

X-FEL R&D (2010-2016)



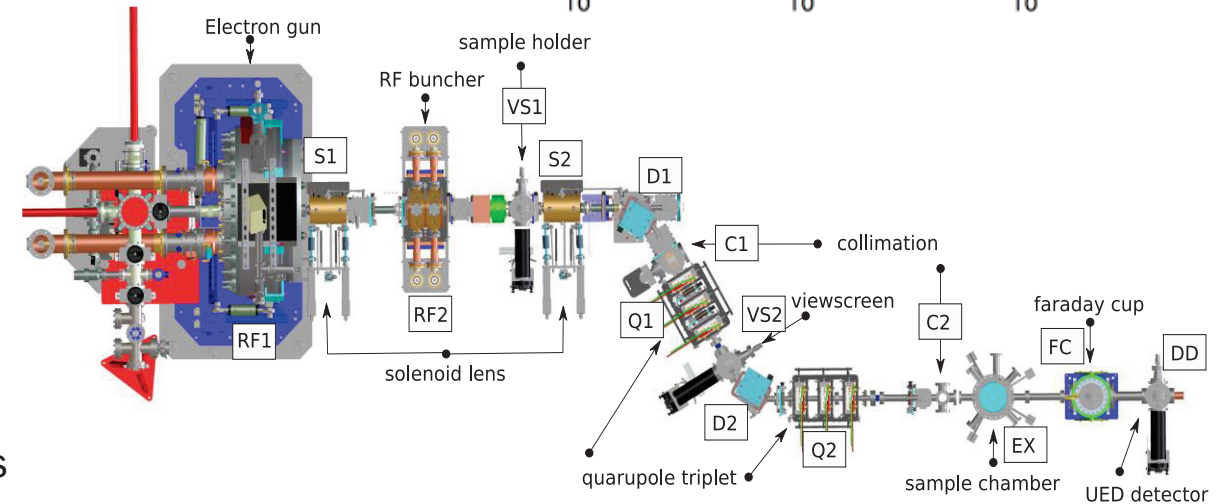
The VHF gun at SLAC
(2016-today)

High Repetition Rate Electron Scattering (HiRES) beamline



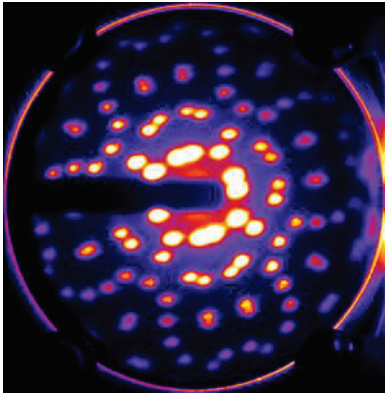
Electron beam energy	700-900 keV
Repetition rate	1-10 ⁶ Hz
Temporal resolution	150 fs-1000 ps
Lateral coherence	1-10 nm
Electrons per pulse	1-10 ⁸
Relative energy spread	10 ⁻³ 10 ⁻⁴

- High rep rate electron source
- High average brightness
- RF-Buncher for longitudinal focusing
- Energy filter
- High lateral coherence
- High flexibility for solid/gas/UED/UEM applications

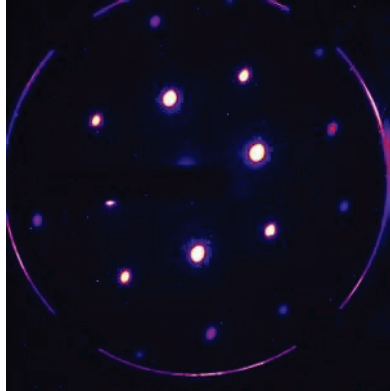


UED tests at HiRES

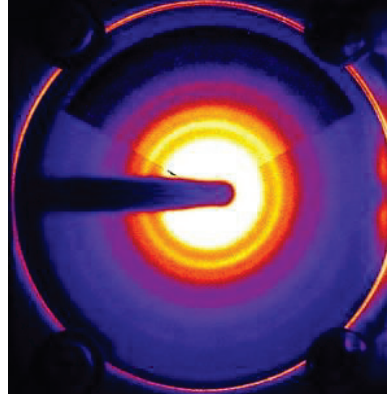
BiXtal gold



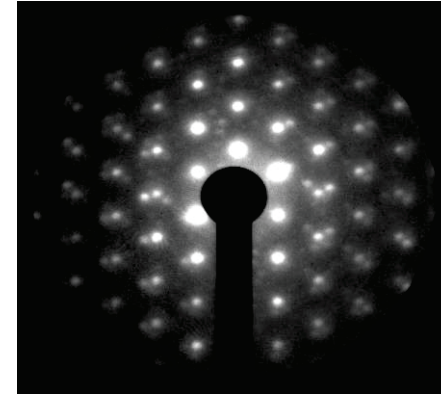
Single Xtal Silicon



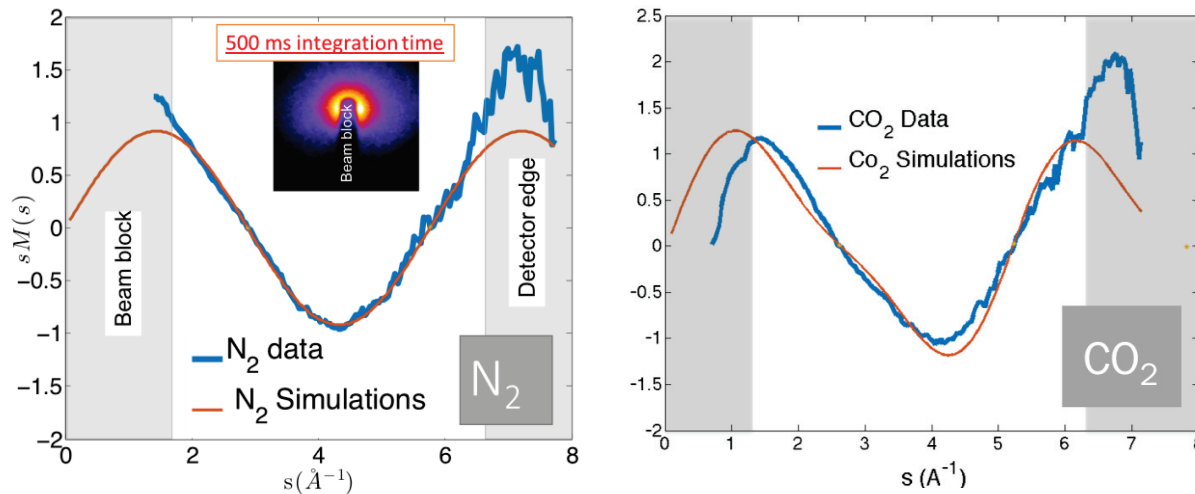
Poly Aluminum



1T – TaS₂

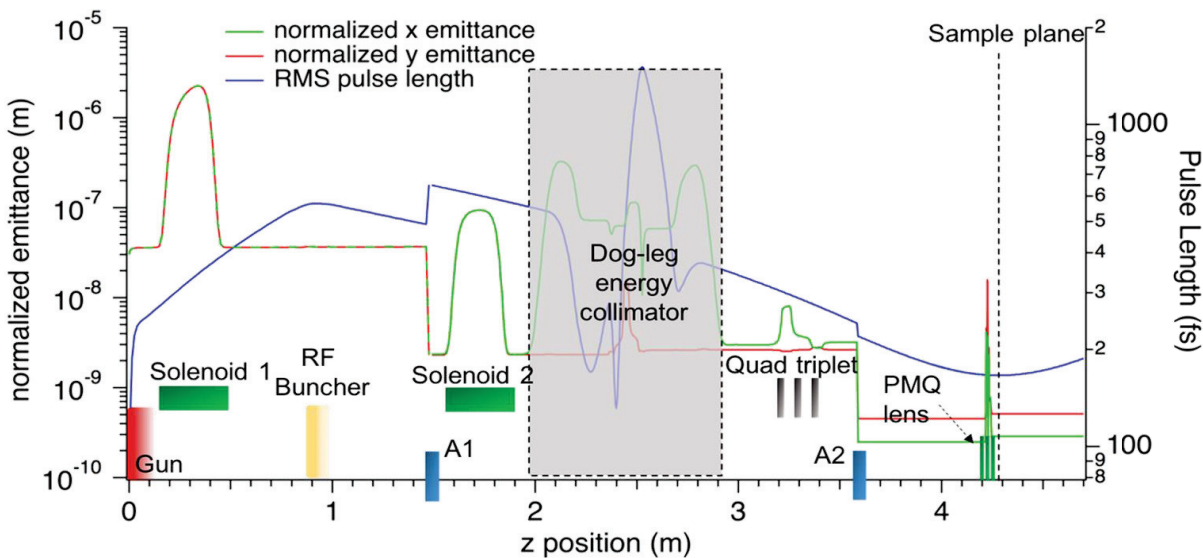
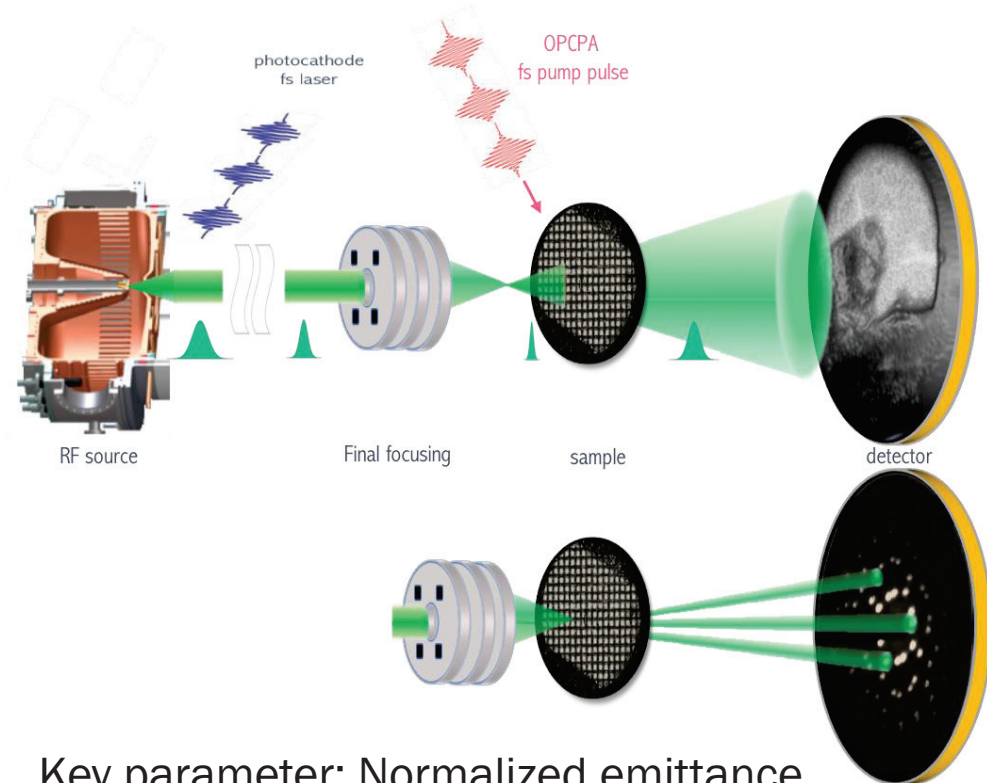


GUED test in 2018



- Solid state test at 10 KHz
- 10 ms Acquisition time
- 3E4 electrons per pulse
- Field of view: $\pm 10 \text{\AA}^{-1}$
- Reciprocal space resolution down to $0.1 \text{\AA}^{-1}$
- Gas phase test at 1MHz
- Pump&Probe test in progress

Ultrafast relativistic electron Nanoprobes



- Key parameter: Normalized emittance

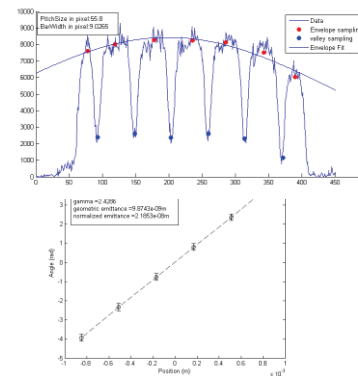
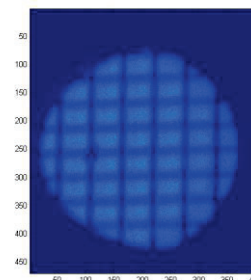
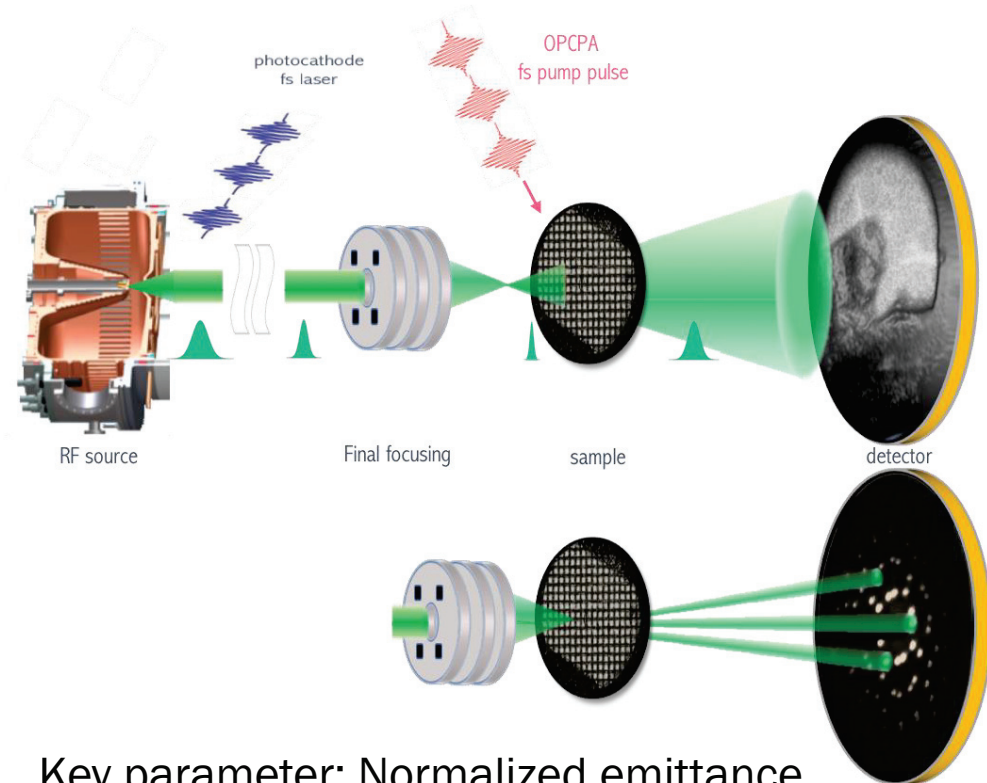
Lateral coherence:
$$L_c = \frac{h}{\sigma_{p\perp}} = \lambda_c \frac{1}{\gamma\beta\sigma_u} = \frac{\lambda_c\sigma_u}{\varepsilon_n}$$

$$L_c = 2 \text{ nm}$$

$$\sigma_u = 500 \text{ nm} \Rightarrow \varepsilon_n \leq 600 \text{ pm}$$

- Short electron pulses generated by the high rep rate RF gun
- RF buncher utilized for longitudinal focusing
- 2 apertures reduce the emittance to the sub-nm regime
- Strongly focused beam for multiple applications: UPPM/USTEM/nanoUED

Ultrafast relativistic electron Nanoprobes



3.06 nrad after the first aperture

In Collaboration with UCLA

$$\epsilon_x = \sqrt{\langle x^2 \rangle \langle x'^2 \rangle - \langle x x' \rangle^2}$$

Phys. Rev. Accel. Beams 21, 10282 (2018)

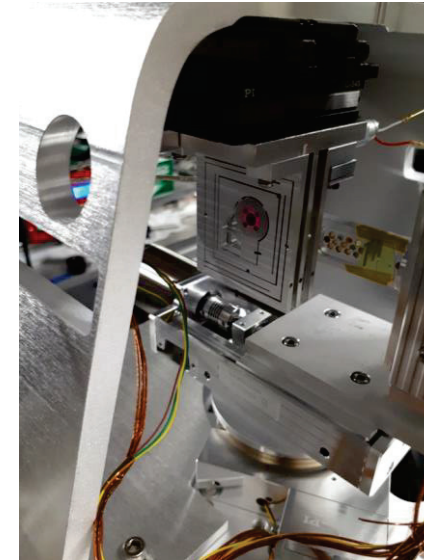
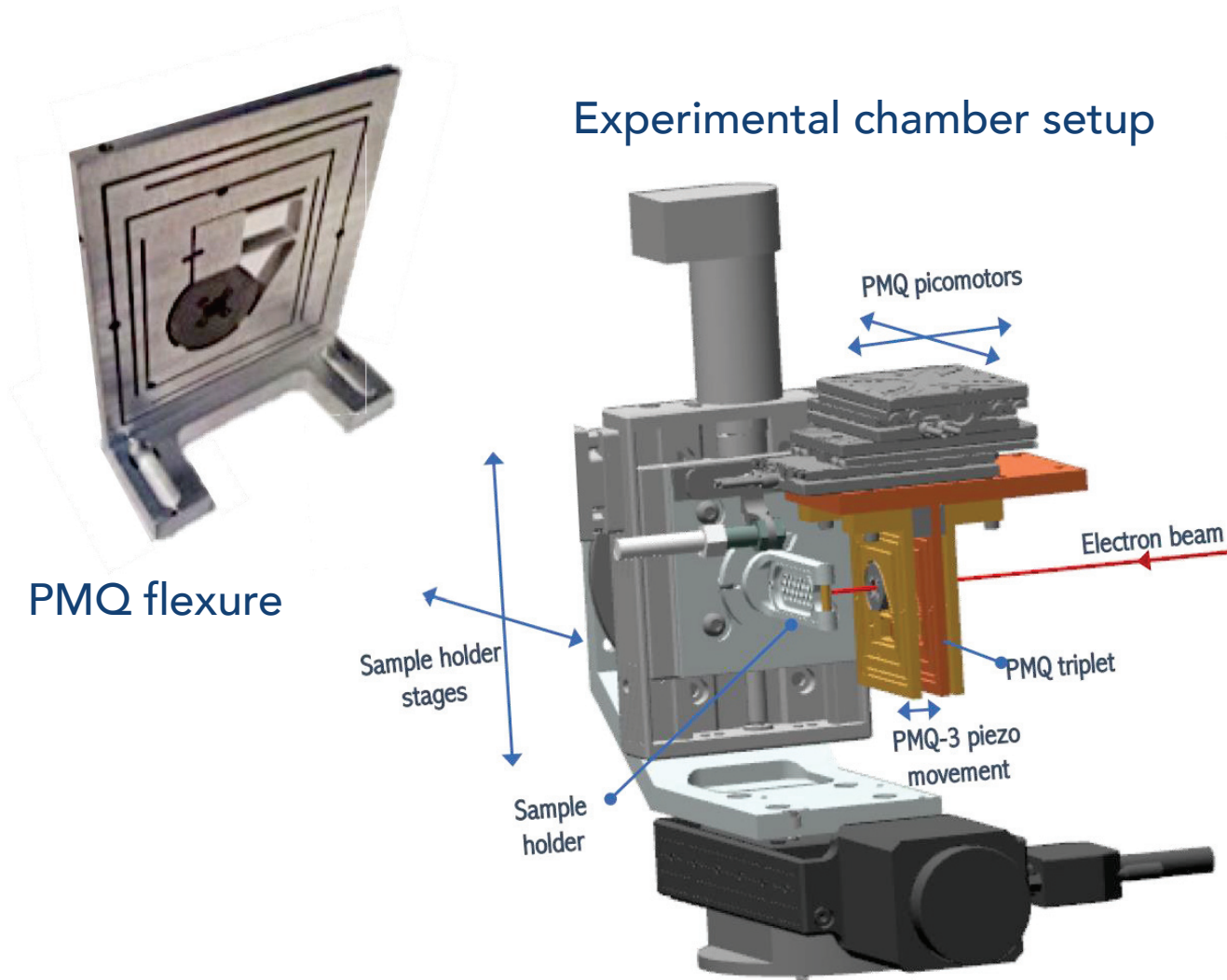
- Key parameter: Normalized emittance

Lateral coherence:
$$L_c = \frac{h}{\sigma_{p\perp}} = \lambda_c \frac{1}{\gamma \beta \sigma_{u'}} = \frac{\lambda_c \sigma_u}{\epsilon_n}$$

$L_c = 2 \text{ nm}$
 $\sigma_u = 500 \text{ nm}$ $\Rightarrow \epsilon_n \leq 600 \text{ pm}$

- 50 um cathode laser, 60nA
- TEM grid diagnostics to measure the beam emittance after 1st aperture
- 3 nm at 320 pA
- Electron optics down stream tuned to partition the beam
- 2nd aperture right before the final focusing optics to further reduce emittance into sub-nm regime

Lateral squeezing of ultrafast electron beam



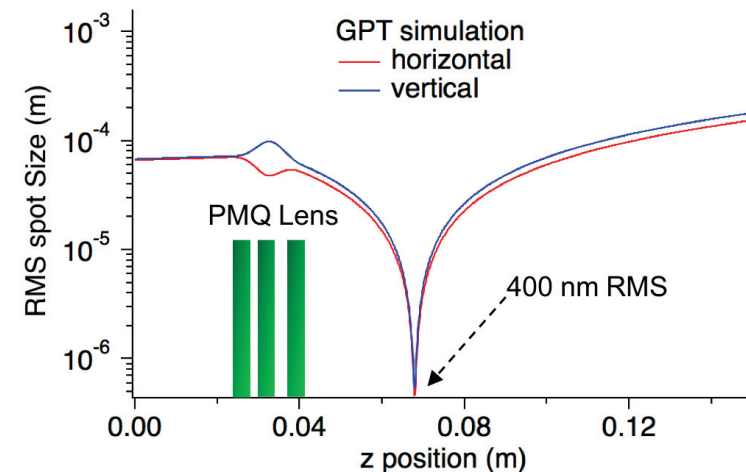
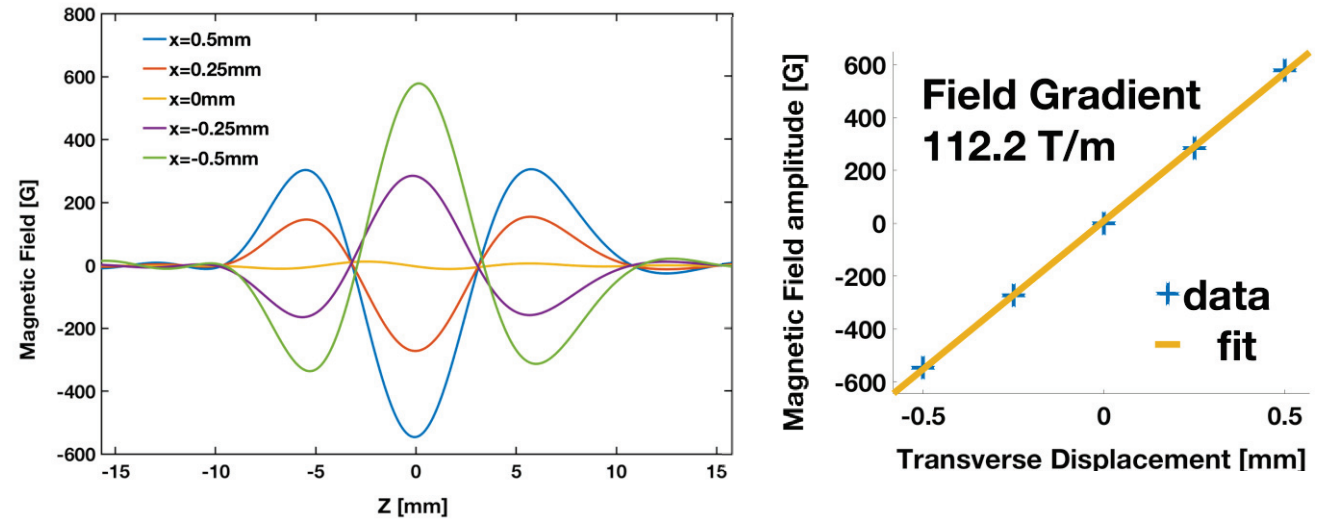
- Home made permanent quadrupole lens with focusing strength $> 100 \text{ T/m}$
- Triplet design meets the lowest requirement of focusing in both direction
- Nominal focal length 25 mm
- Step motor, remote control during operation
- 3rd quadrupole flexure is adjustable to fine tune the horizontal and vertical focal point
- All in vacuum, UHV compatible

Lateral squeezing of ultrafast electron beam

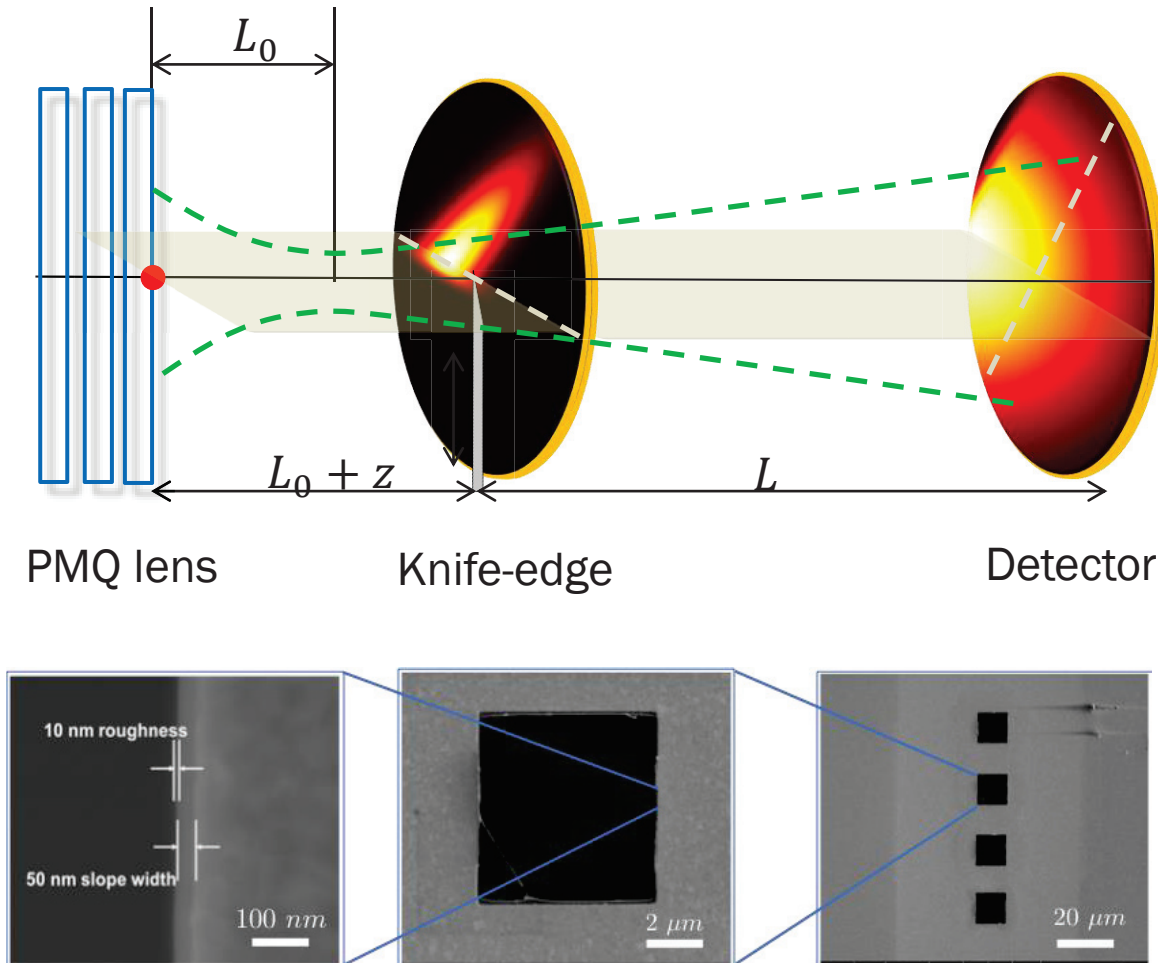


Nuclear Instruments and Methods in Physics Research A 716 (2013) 62–70

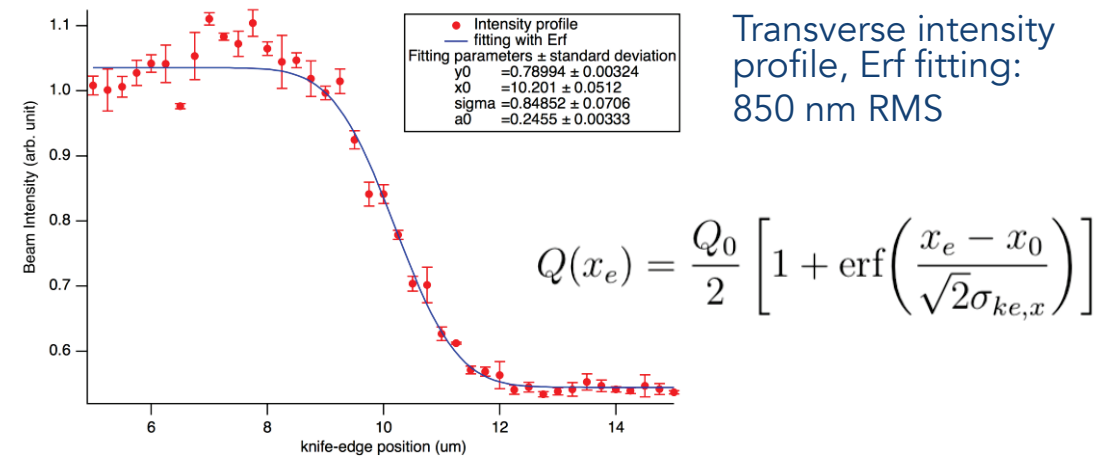
- Pulsed wire set up in the engineering division, LBNL
- PMQ alignment better than 15 μm
- Field gradient $> 100 \text{ T/m}$
- Simulations shows 400 nm RMS spot size can be achieved with 600 pm emittance



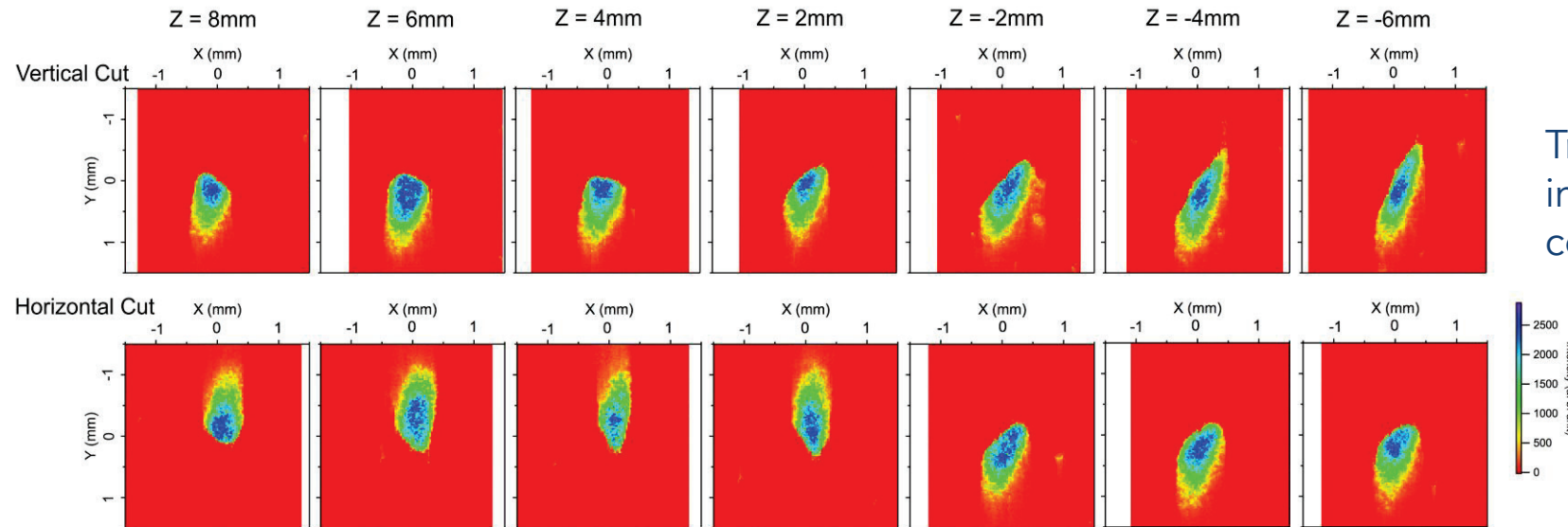
Sub - um electron beam characterization



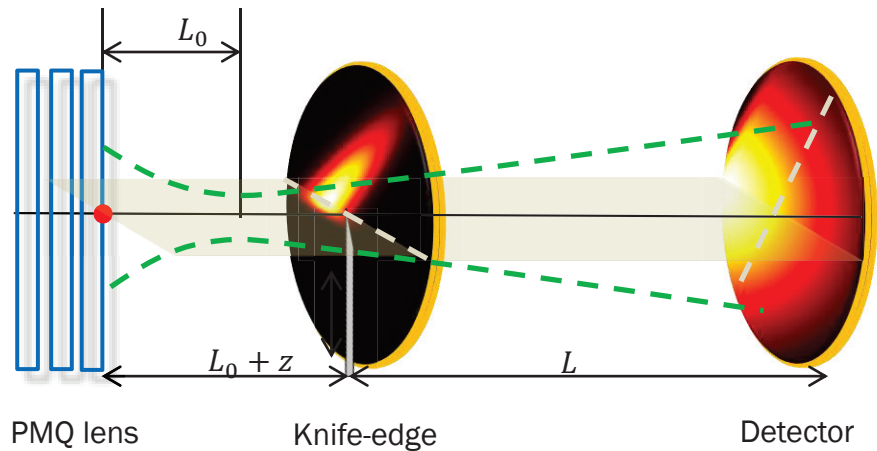
- Characterization of sub-um electron beam is an active research field
- Knife edge target fabricated using focused ion beam (FIB) in Molecular Foundry/LBNL
- Gradually inserted into the beam to get projected beam size in horizontal/vertical
- Quadrupole scan/Longitudinal beam size scan only gives the 2D projected beam emittance
- How to extract **uncorrelated emittance** from the measurement?



Sub - um electron beam characterization



Transmitted Beam profile after intercepting the knife-edge in the center



- Cut angle rotates along the longitudinal scan
- There is a correlation between x and y at the knife-edge plane
- Need to obtain the full **4D beam matrix** to understand the beam evolution, but how?

Theoretical model

4D beam matrix in the general case

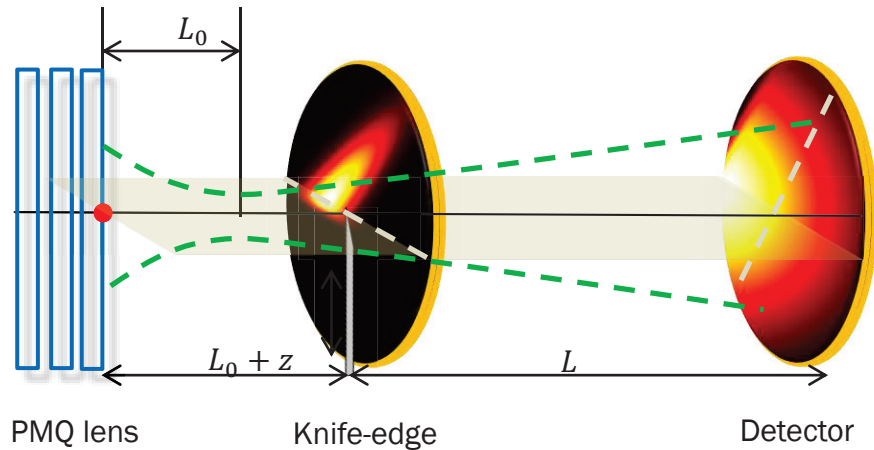
$$\Sigma_{4D} = \begin{pmatrix} \langle x^2 \rangle & \langle xx' \rangle & \langle xy \rangle & \langle xy' \rangle \\ \langle xx' \rangle & \langle x'^2 \rangle & \langle x'y \rangle & \langle x'y' \rangle \\ \langle xy \rangle & \langle x'y \rangle & \langle y^2 \rangle & \langle yy' \rangle \\ \langle xy' \rangle & \langle x'y' \rangle & \langle yy' \rangle & \langle y'^2 \rangle \end{pmatrix}$$

4D emittance

$$\epsilon_{4D} = \sqrt{\det(\Sigma_{4D})}$$

distribution limit at the detector plane

$$\rho(x, x', y, y') = \begin{cases} \frac{Q_0}{4\pi^2 \epsilon_{4D}} \exp \left[-\frac{1}{2} \begin{pmatrix} x & x' & y & y' \end{pmatrix} \Sigma_{4D}^{-1} \begin{pmatrix} x \\ x' \\ y \\ y' \end{pmatrix} \right], & \text{if } x > x_e + Lx' \\ 0, & \text{if } x \leq x_e + Lx' \end{cases}$$



Beam profile on the detector

$$\begin{aligned} \rho_x(x, y) &= \int_{-\infty}^{\frac{x-x_e}{L}} dx' \int_{-\infty}^{\infty} dy' \rho(x, x', y, y') \\ &= \frac{1}{2} \rho_0(x, y) \left[1 + \operatorname{erf} \left(\frac{x - M_x x_e - \tan(\alpha_x) y}{\sqrt{2} \sigma_x} \right) \right] \end{aligned}$$

Theoretical model

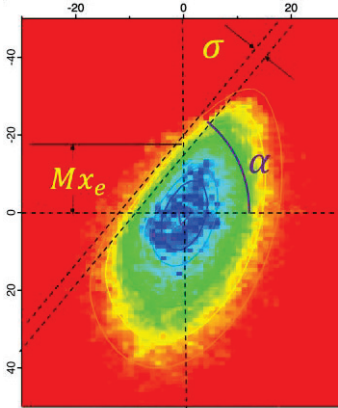
Beam profile on the detector

Horizontal Knife-edge

$$\begin{aligned}\rho_x(x, y) &= \int_{-\infty}^{\frac{x-x_e}{L}} dx' \int_{-\infty}^{\infty} dy' \rho(x, x', y, y') \\ &= \frac{1}{2} \rho_0(x, y) \left[1 + \operatorname{erf} \left(\frac{x - M_x x_e - \tan(\alpha_x) y}{\sqrt{2} \sigma_x} \right) \right]\end{aligned}$$

Vertical Knife-edge

$$\rho_y(x, y) = \frac{1}{2} \rho_0(x, y) \left[1 + \operatorname{erf} \left(\frac{y - M_y y_e - \tan(\alpha_y) x}{\sqrt{2} \sigma_y} \right) \right]$$



4D beam matrix at the knife-edge plane

$$\Sigma_{4D}^{-1} = \begin{pmatrix} a & b & g & h \\ b & c & l & m \\ g & l & d & e \\ h & m & e & f \end{pmatrix}$$

Fitting parameters related to the beam matrix elements

Transverse profile without the knife-edge

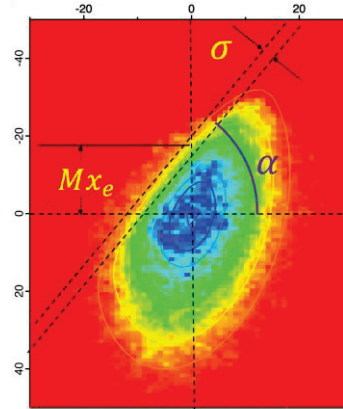
$$\begin{aligned}\rho_0(x, y) &= \frac{Q_0}{2\pi \sqrt{\langle x^2 \rangle \langle y^2 \rangle - \langle xy \rangle^2}} \times \\ &\exp \left[-\frac{1}{2} \begin{pmatrix} x - x_0 & y - y_0 \end{pmatrix} \begin{pmatrix} \langle x^2 \rangle & \langle xy \rangle \\ \langle xy \rangle & \langle y^2 \rangle \end{pmatrix}^{-1} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \right]\end{aligned}$$

$$\begin{aligned}M_x &= \frac{1}{1 + L \frac{bf - hm}{cf - m^2}} & M_y &= \frac{1}{1 + L \frac{ce - lm}{cf - m^2}} \\ \tan(\alpha_x) &= \frac{em - fl}{bf - hm + \frac{cf - m^2}{L}} & \tan(\alpha_y) &= \frac{bm - ch}{ce - lm + \frac{cf - m^2}{L}} \\ \sigma_x &= \frac{\sqrt{f(cf - m^2)}}{bf - hm + \frac{cf - m^2}{L}} & \sigma_y &= \frac{\sqrt{c(cf - m^2)}}{ce - lm + \frac{cf - m^2}{L}}\end{aligned}$$

Theoretical model

Fitting parameters related to the beam matrix elements

$$\begin{aligned}
 M_x &= \frac{1}{1 + L \frac{bf - hm}{cf - m^2}} & M_y &= \frac{1}{1 + L \frac{ce - lm}{cf - m^2}} \\
 \tan(\alpha_x) &= \frac{em - fl}{bf - hm + \frac{cf - m^2}{L}} & \tan(\alpha_y) &= \frac{bm - ch}{ce - lm + \frac{cf - m^2}{L}} \\
 \sigma_x &= \frac{\sqrt{f(cf - m^2)}}{bf - hm + \frac{cf - m^2}{L}} & \sigma_y &= \frac{\sqrt{c(cf - m^2)}}{ce - lm + \frac{cf - m^2}{L}}
 \end{aligned}$$



4D beam matrix at the knife-edge plane

$$\Sigma_{4D}^{-1} = \begin{pmatrix} a & b & g & h \\ b & c & l & m \\ g & l & d & e \\ h & m & e & f \end{pmatrix}$$

10 parameter Global fitting to minimize the systematic/statistical error

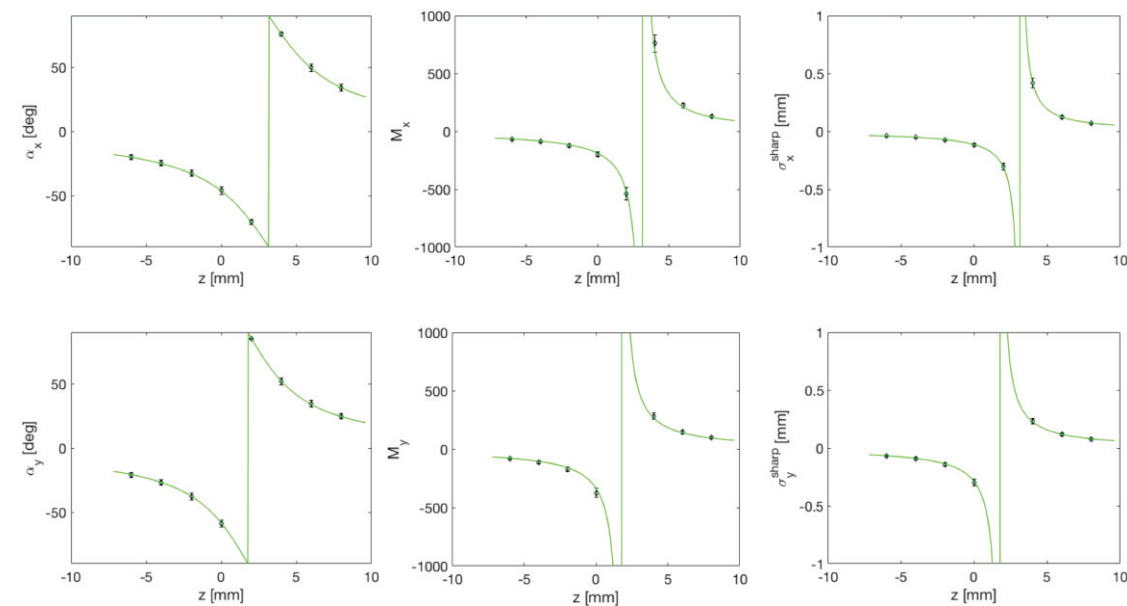
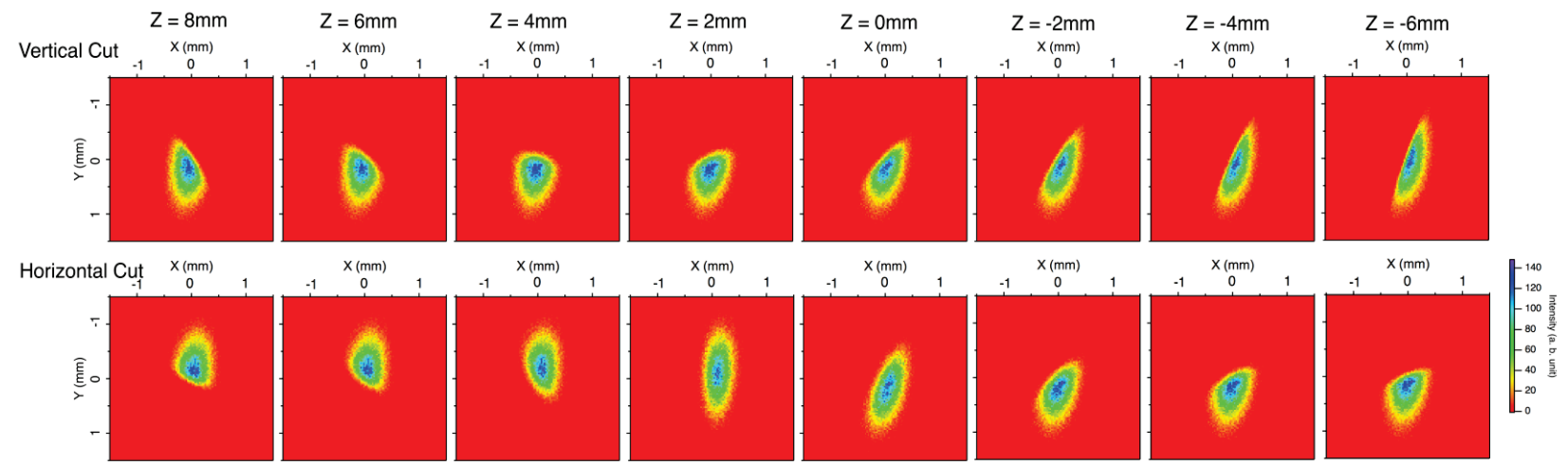
Matlab routine for minimization of the squared error between the data s_i and the parametric model t_i

$$\chi^2 \left(\Sigma_{4D}^{(\text{recon})}, L_0 \right) = \sum_{k,i} \left(\frac{s_i^{(k)} - t_i^{(k)}(\Sigma_{4D}^{(\text{recon})}, L_0)}{\delta s_i^{(k)}} \right)^2$$

- Dependence of the knife-edge to screen distance L
- In principle, all beam matrix elements can be retrieved from M , $\tan(\alpha)$, σ and beam size at knife edge for horizontal and vertical scans
- In practice, ...

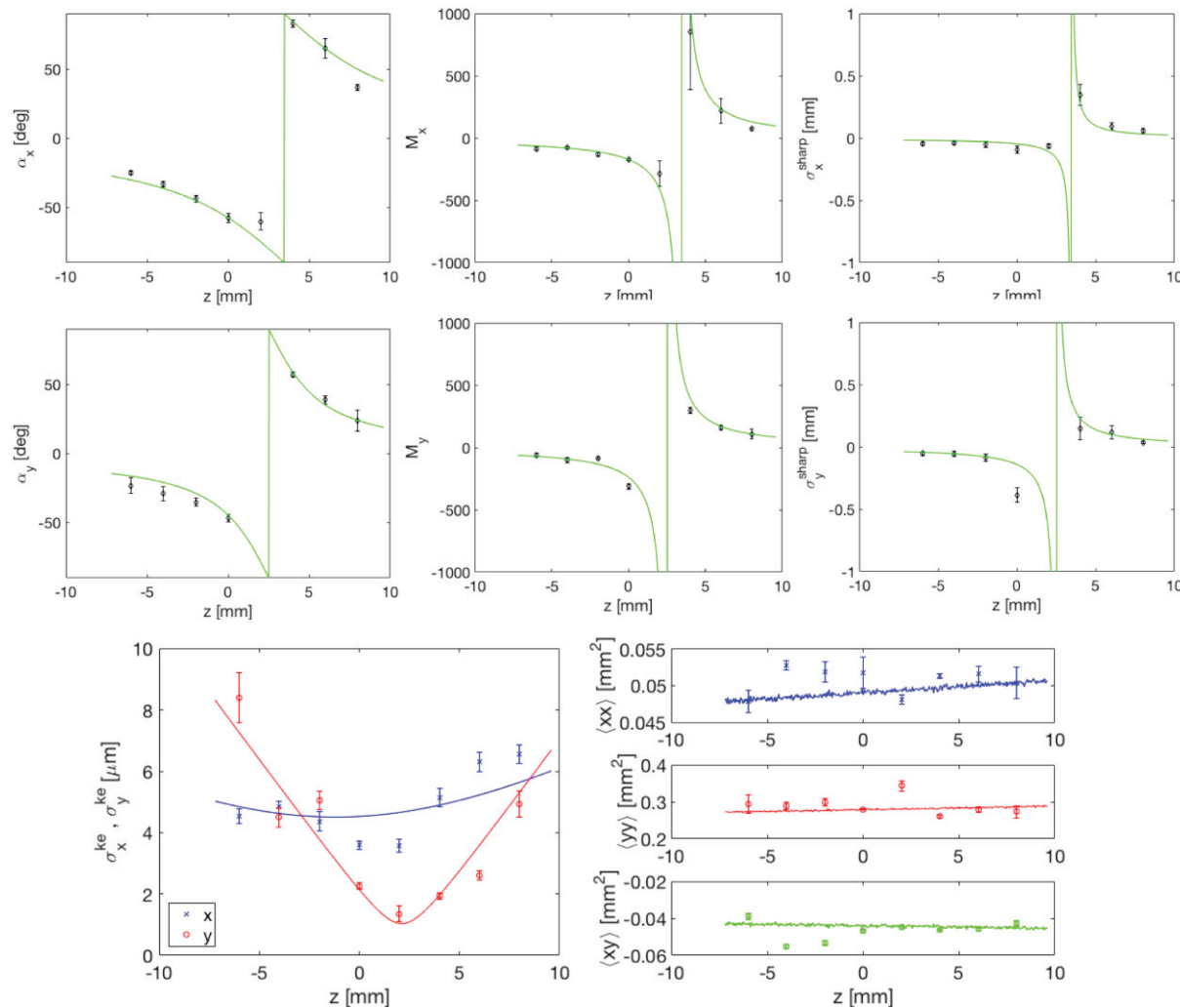
Virtual measurements

		Initial	Reconstructed
$\langle x^2 \rangle$	$[\mu\text{m}^2]$	59.83	59.20
$\langle xx' \rangle$	$[\mu\text{m mrad}]$	-2.432	-2.406
$\langle x'^2 \rangle$	$[\text{mrad}^2]$	0.1087	0.1079
$\langle y^2 \rangle$	$[\mu\text{m}^2]$	256.5	256.4
$\langle yy' \rangle$	$[\mu\text{m mrad}]$	-11.38	-11.37
$\langle y'^2 \rangle$	$[\text{mrad}^2]$	50.76	50.69
$\langle xy \rangle$	$[\mu\text{m}^2]$	11.27	11.18
$\langle xy' \rangle$	$[\mu\text{m mrad}]$	-0.2045	-0.2085
$\langle x'y \rangle$	$[\mu\text{m mrad}]$	1.065	1.074
$\langle x'y' \rangle$	$[\text{mrad}^2]$	-0.06062	-0.06082
$\epsilon_{4D}(\text{geometric})$	$[(\text{nm rad})^2]$	0.001950	0.001948
$\epsilon_{4D}(\text{normalized})$	$[(\text{nm rad})^2]$	0.009642	0.009632



- Knife-edge scan in GPT simulation
- The extracted values are fed into the global fitting routine
- Excellent agreement of initial and reconstructed matrix elements
- 4D emittance is extremely small: $0.009 (\text{nm rad})^2$

Global fitting of the measured data

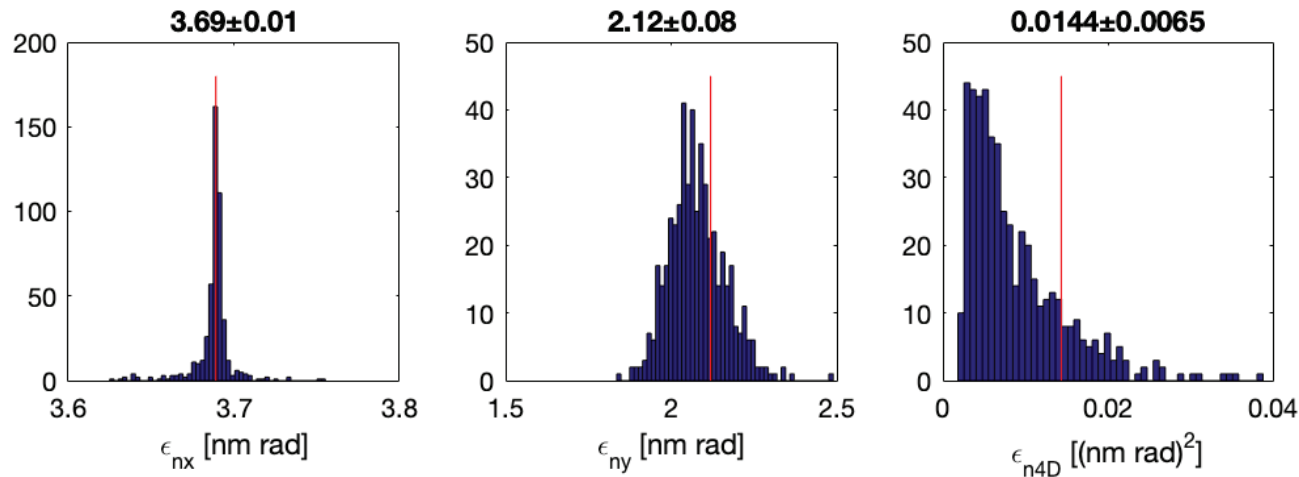


$\langle x^2 \rangle$	[μm^2]	73.4 ± 0.3
$\langle xx' \rangle$	[$\mu\text{m mrad}$]	-2.69 ± 0.01
$\langle x'^2 \rangle$	[mrad^2]	0.1365 ± 0.0004
$\langle y^2 \rangle$	[μm^2]	420 ± 0.9
$\langle yy' \rangle$	[$\mu\text{m mrad}$]	-18.14 ± 0.02
$\langle y'^2 \rangle$	[mrad^2]	0.786 ± 0.002
$\langle xy \rangle$	[μm^2]	40.30 ± 0.08
$\langle xy' \rangle$	[$\mu\text{m mrad}$]	-1.42 ± 0.01
$\langle x'y \rangle$	[$\mu\text{m mrad}$]	2.38 ± 0.01
$\langle x'y' \rangle$	[mrad^2]	-0.1161 ± 0.0003
$\epsilon_{4D}(\text{geometric})$	[$(\text{nm rad})^2$]	0.0029 ± 0.0013
$\epsilon_{4D}(\text{normalized})$	[$(\text{nm rad})^2$]	0.0144 ± 0.0065

- Optimization weight: standard deviation of the data points
- Full 4D beam matrix retrieved
- Extremely small 4D normalized emittance, $0.0144 (\text{nm rad})^2$

Error analysis

Monte Carlo simulations



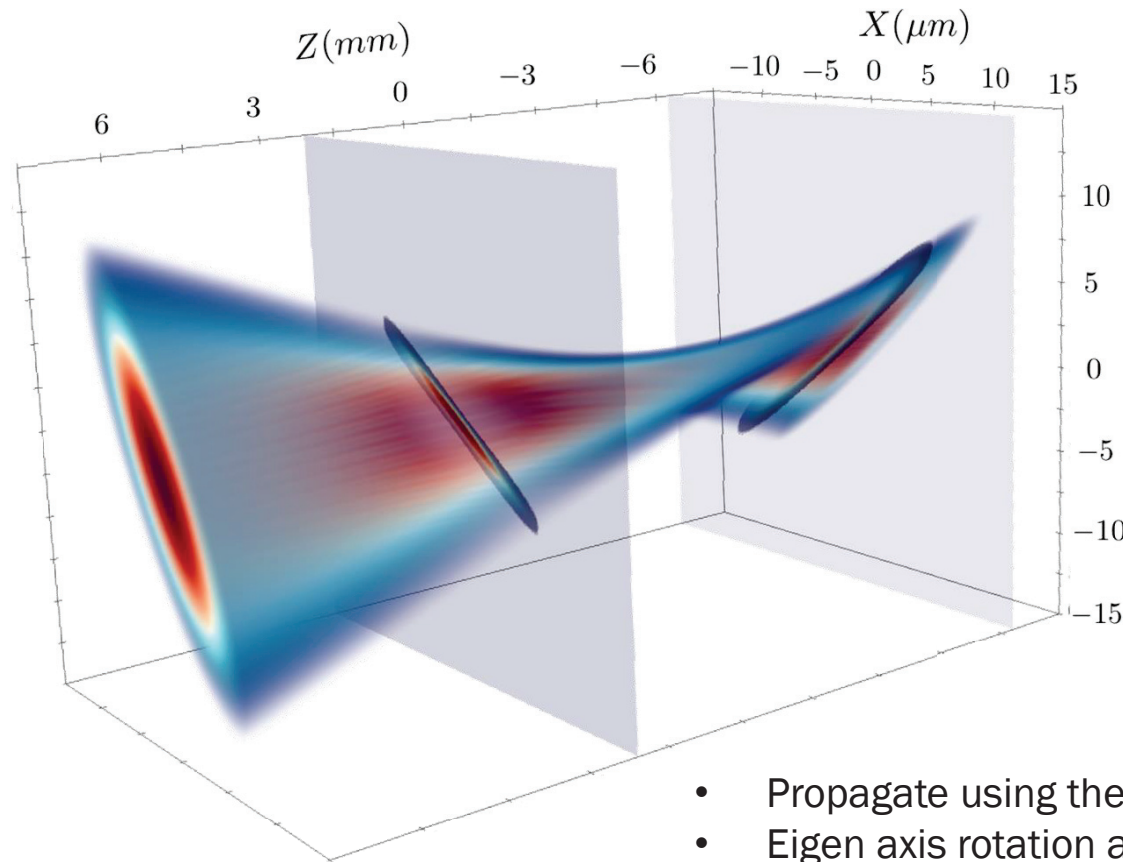
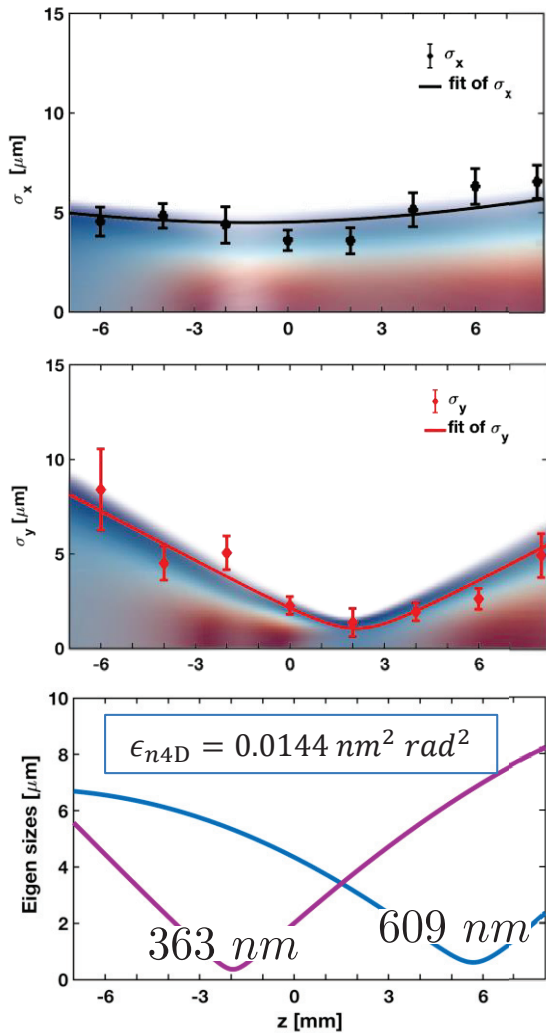
- 5000 data sets randomly generated based on their profile and uncertainties
- Every data set is processed as input of the global fitting
- Obtaining statistical distribution of the emittance values
- Two orders of magnitude difference between the 4D and projected emittances -> confirms the x-y correlation

$\langle x^2 \rangle$	[μm^2]	73.4 ± 0.3
$\langle xx' \rangle$	[$\mu\text{m mrad}$]	-2.69 ± 0.01
$\langle x'^2 \rangle$	[mrad^2]	0.1365 ± 0.0004
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$\langle x'y' \rangle$	[mrad^2]	-0.1161 ± 0.0003

$\epsilon_{4D}(\text{geometric})$	[(nm rad) ²]	0.0029 ± 0.0013
$\epsilon_{4D}(\text{normalized})$	[(nm rad) ²]	0.0144 ± 0.0065

- Optimization weight: standard deviation of the data points
- Full 4D beam matrix retrieved
- Extremely small 4D normalized emittance, 0.0144 (nm rad)²

Reconstruct the beam evolution around waist

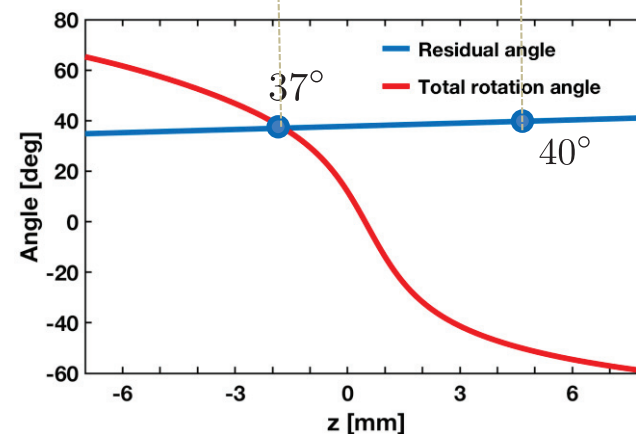
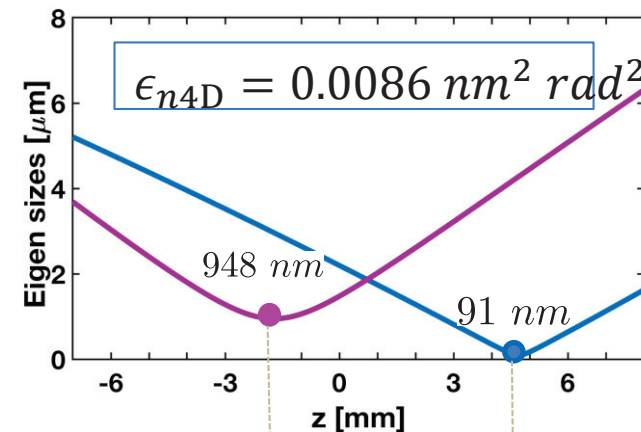
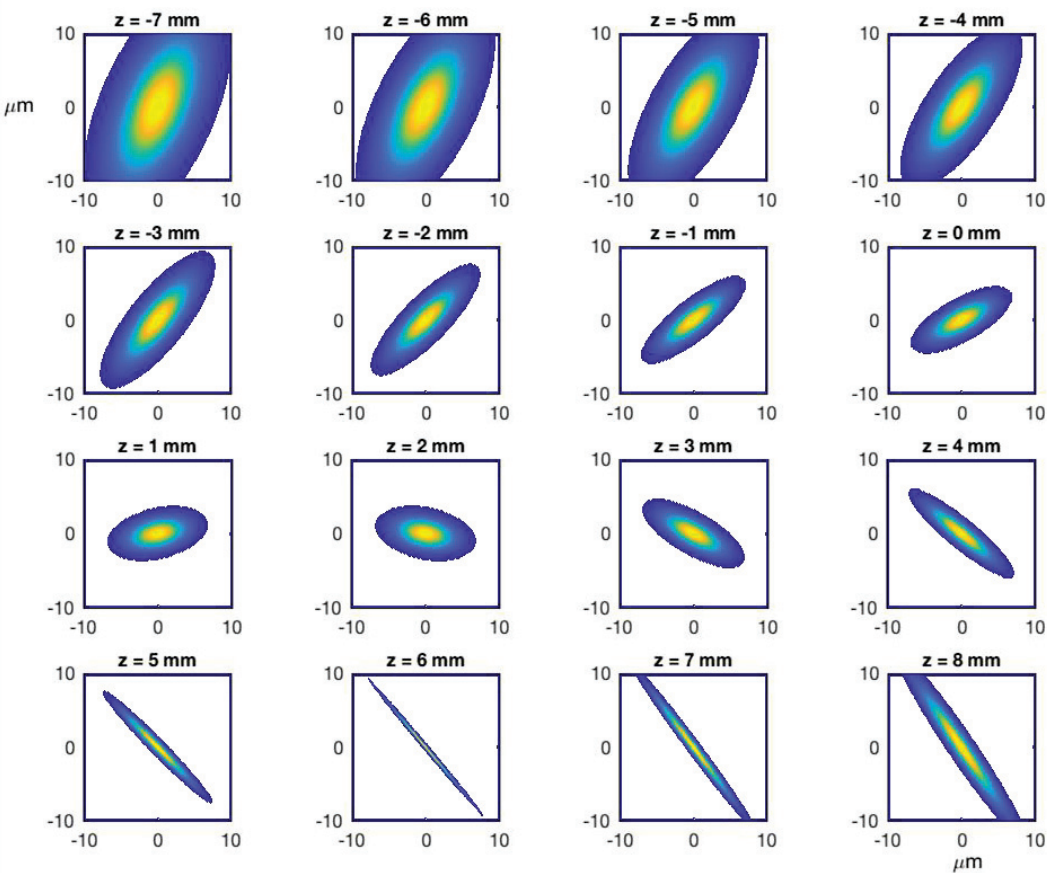


$\langle x^2 \rangle$	[μm ²]	73.4 ± 0.3
$\langle xx' \rangle$	[μm mrad]	-2.69 ± 0.01
$\langle x'^2 \rangle$	[mrad ²]	0.1365 ± 0.0004
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$\epsilon_{4D}(\text{geometric})$	[(nm rad) ²]	0.0029 ± 0.0013
$\epsilon_{4D}(\text{normalized})$	[(nm rad) ²]	0.0144 ± 0.0065

Transverse coherence: $\geq 2.4 \text{ nm}$

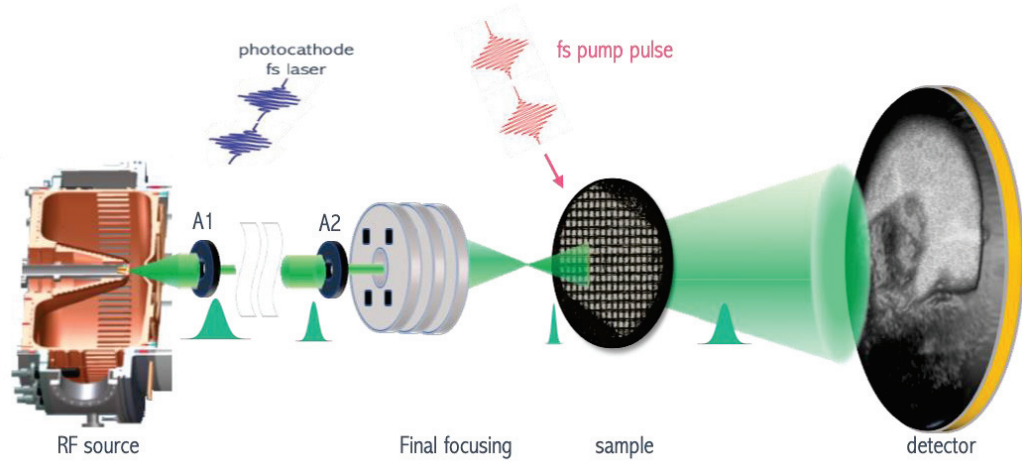
- Propagate using the reconstructed 4D beam matrix
- Eigen axis rotation across the waist
- Eigen size minima as small as 363 nm
- Projection in x/y plane consistent with the measured data points

Reconstruct the beam evolution around waist

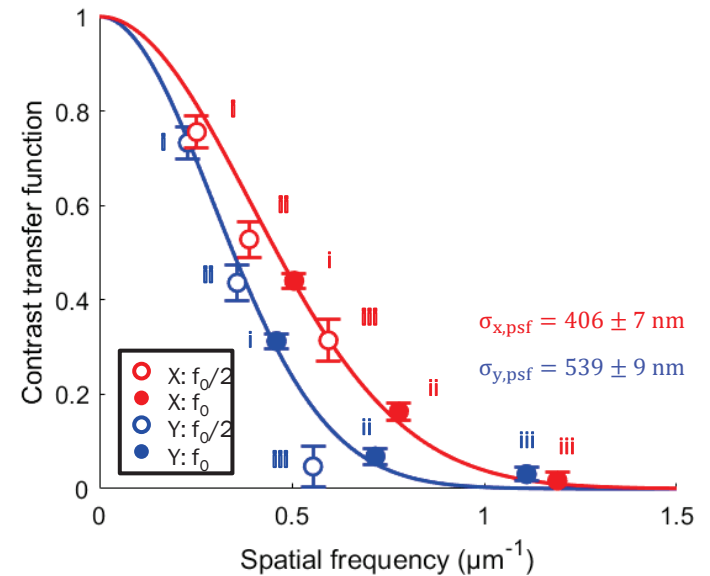
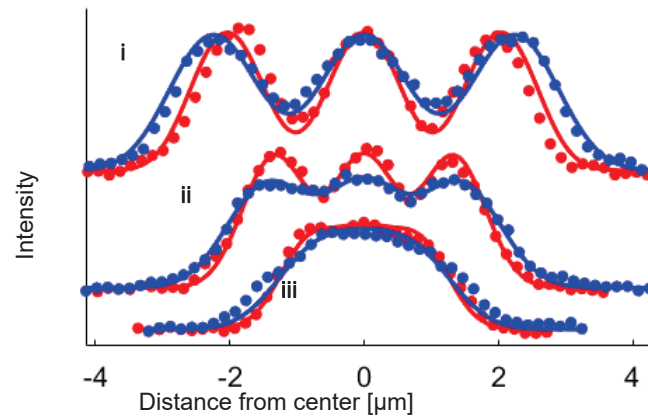
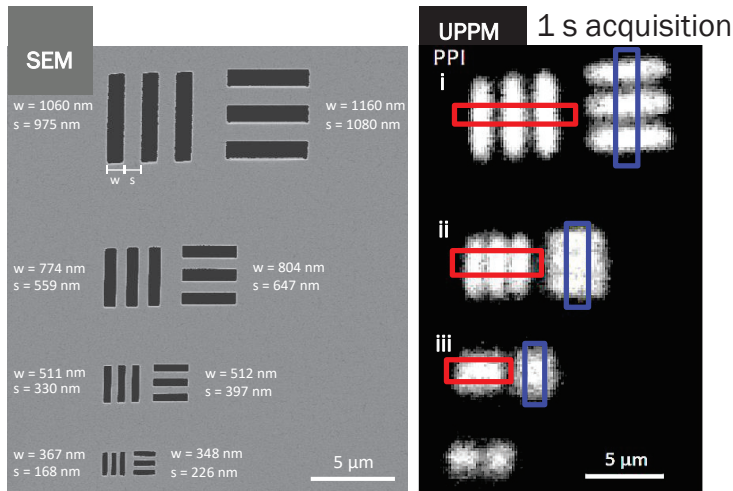


- Partition the beam using upstream electron optics
- 91 nm rms in one direction achieved
- Could be useful in situations where high resolution is only needed in one dimension
- Eigen axis rotation could be induced by angular momentum introduced by a rolling alignment error of the PMQ lens, laser spot astigmatism, etc

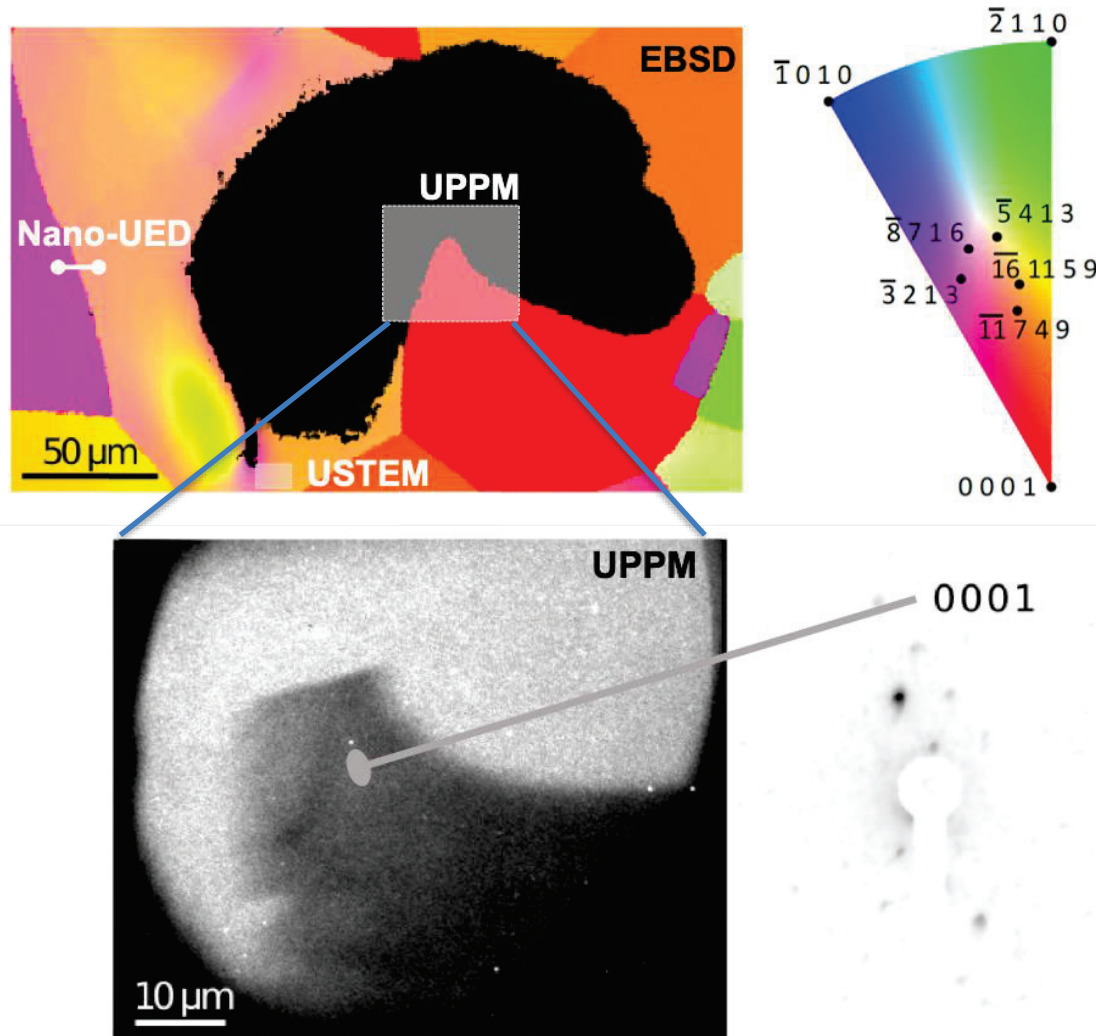
Ultrafast point-projection microscopy



- Focusing plane placed upstream of the sample
- The image on detector represents a magnified mass-contrast shadowgraph of the specimen
- FIB fabricated USAF test chart
- Contrast transfer function (CTF) of spatial frequency

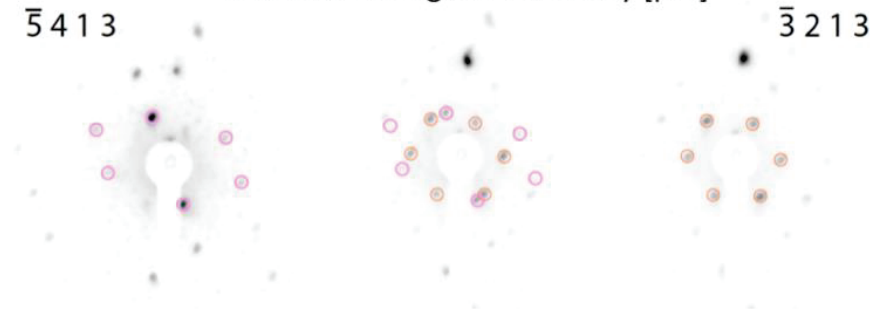
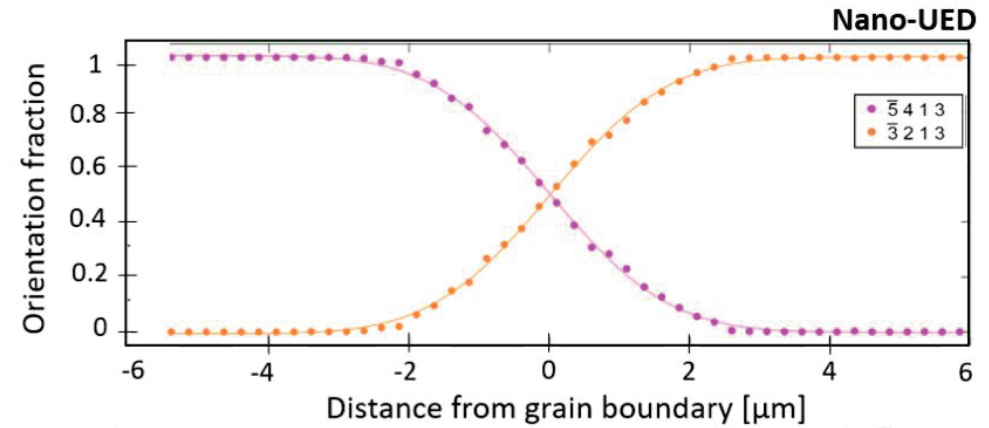
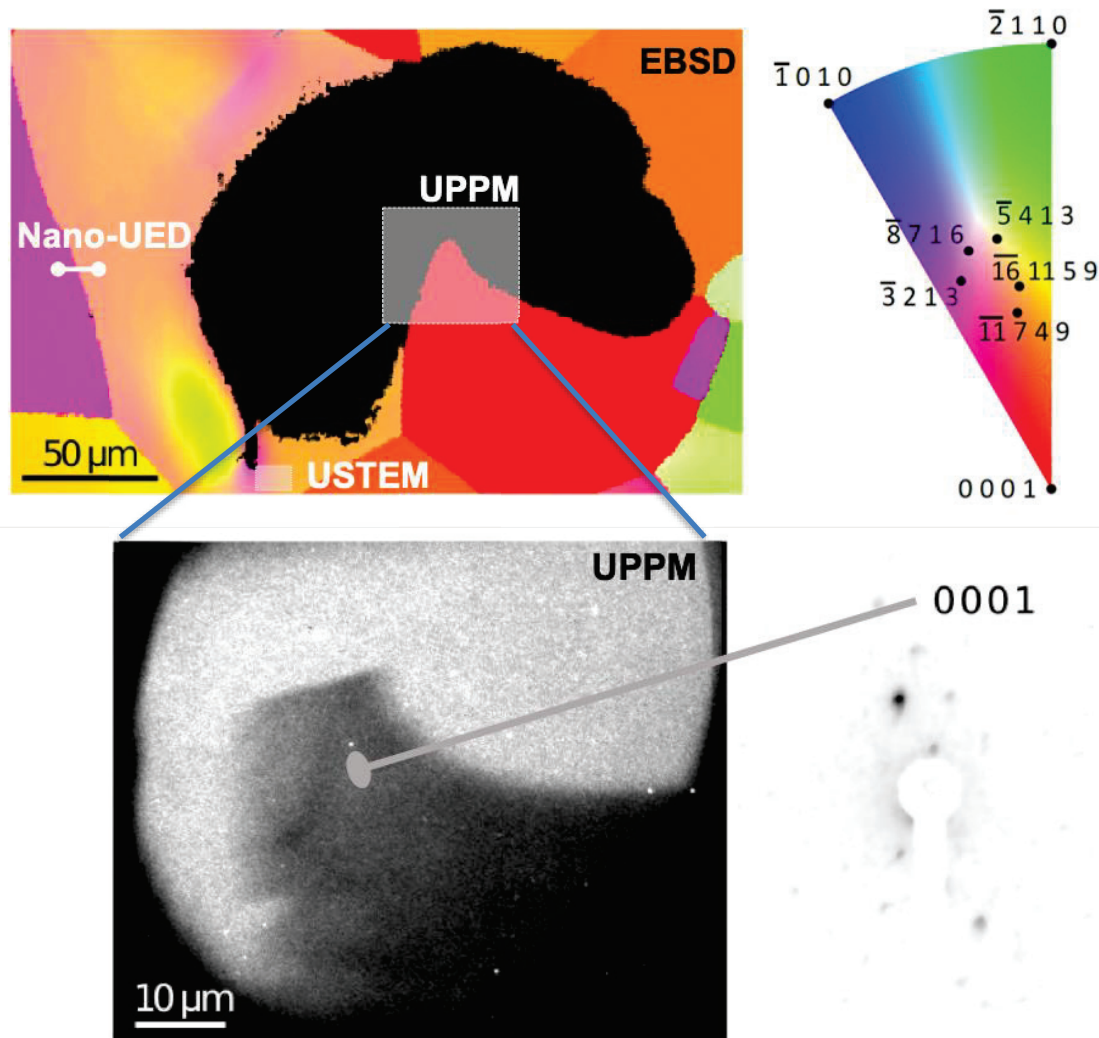


Ultrafast scanning nano-diffraction



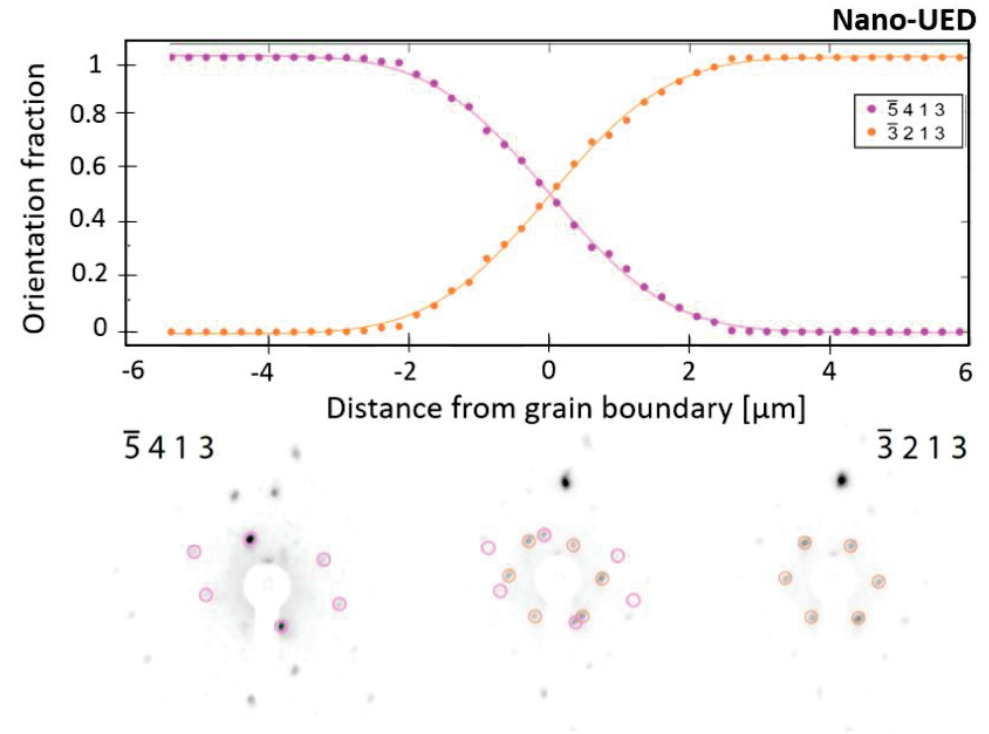
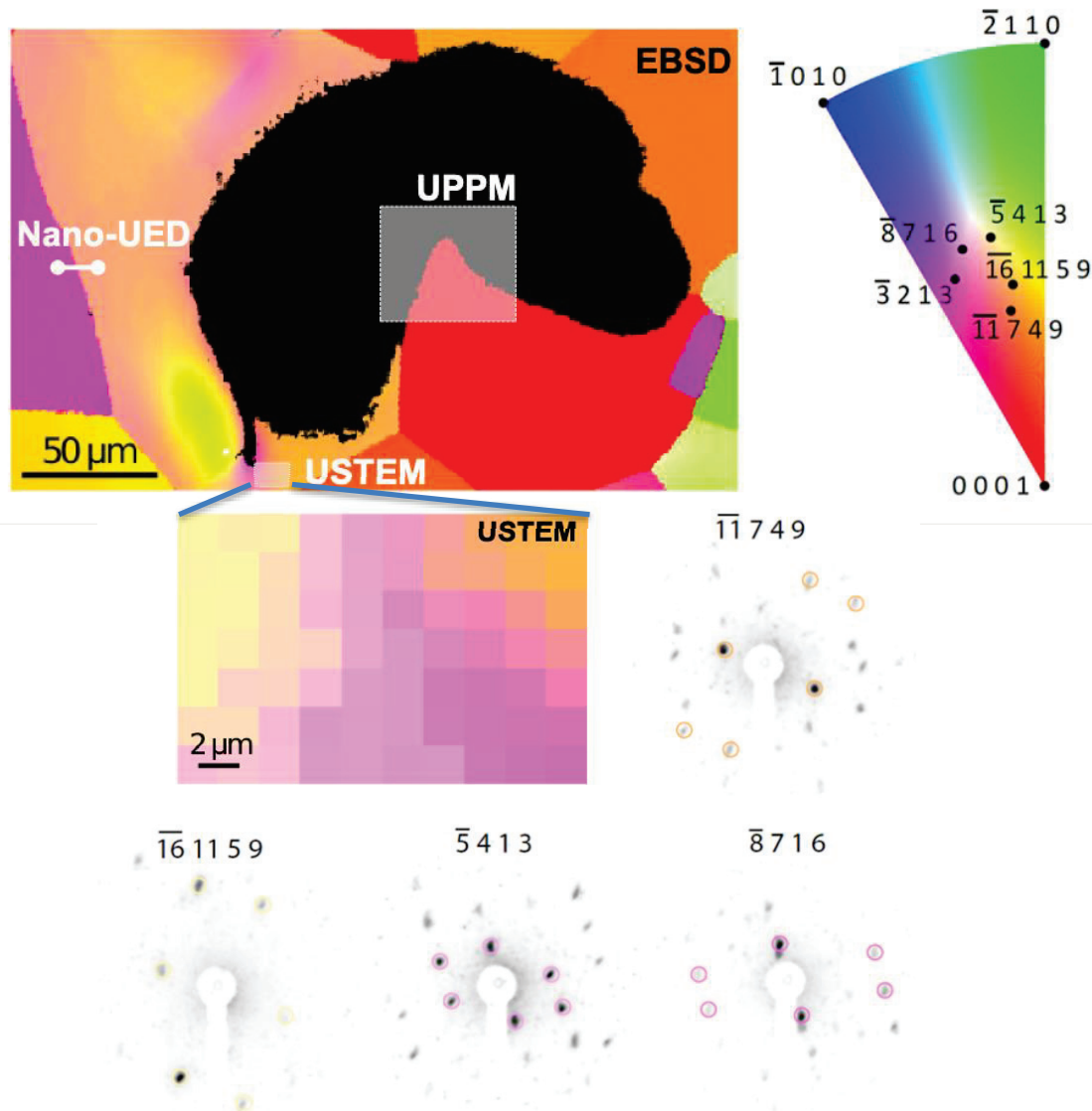
- Example measurement: Ti -6 wt % Al polycrystal
- Utilize UPPM to locate region of interest on sample and
- Switch to nano-UED mode: Focus the beam onto the protrusion and obtain diffraction pattern indicating $[0001]$ orientation
- Matching the orientation determined by EBSD
- Online operations at 1 MHz $\rightarrow$ quick sample locating and data taking

Ultrafast scanning nano-diffraction



- Scanning across the grain boundary with 250 nm steps
- Orientation fraction function obtained from intensities of selected diffraction spots
- Determining the interface position with 15 nm standard error
- Useful for precise nano-UED studies of interfaces/ domain boundaries

Ultrafast scanning nano-diffraction



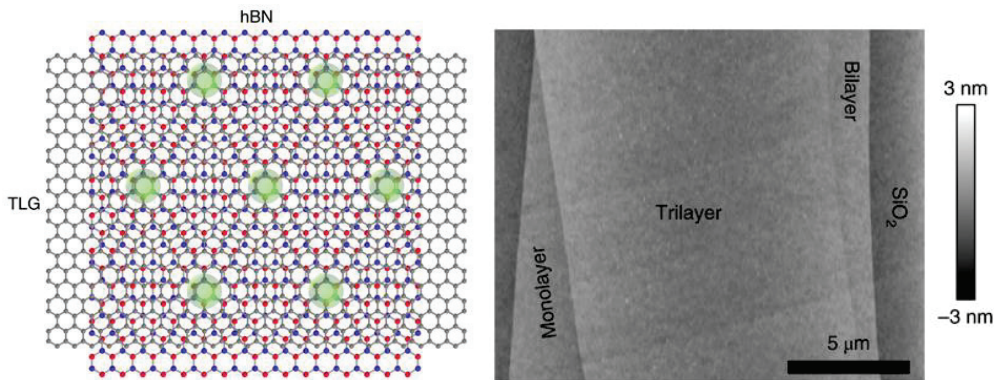
- Scanning transmission electron microscopy (STEM) by mapping the crystal orientation in a 2D region
- Orientation determined using diffraction patterns
- Color map generated using orientation fraction
- A gradual bend observed matching the EBSD map
- Spatial-temporal mapping with sub- μm resolution

Discussion and further works

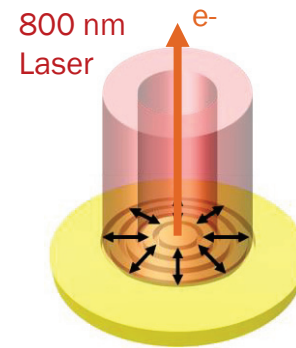
- Relativistic electron probes with size in the sub- μm regime are smaller than typical grain sizes in most materials
- Access to local structure information with fs temporal resolution
- Structure dynamics in nano-wires, nano-particles and small grains of exfoliated/epitaxial low dimensional materials
- Coherent phonon excitation in TMDC micro-crystallites
- Electron-phonon coupling in functional Morrie pattern heterostructures

- High brightness, low emittance electron source R&D
- Plasmonic photocathodes
- Beamline optimization
- Temporal jitter studies
- Single shot time stamping

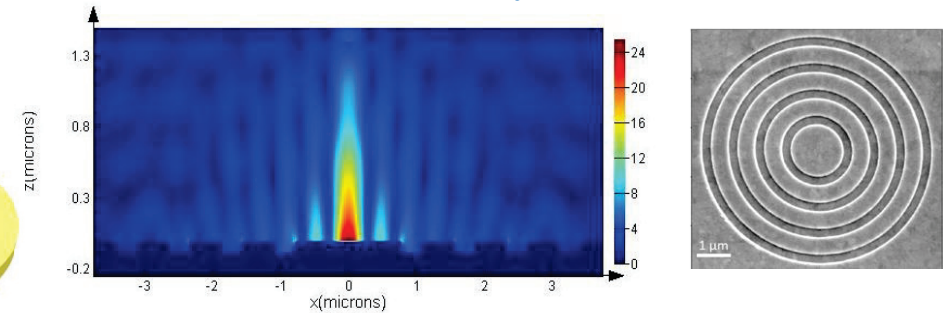
Collab: UCB Feng Wang



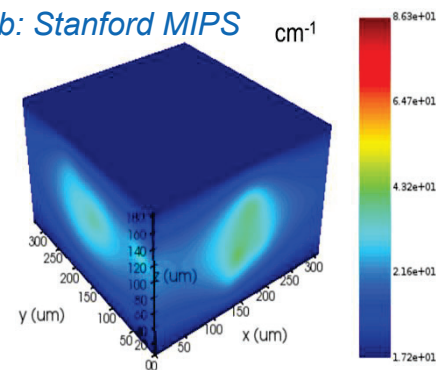
Nature Physics 15, 237–241 (2019)



Collab: LBNL Molecular Foundry, UCLA

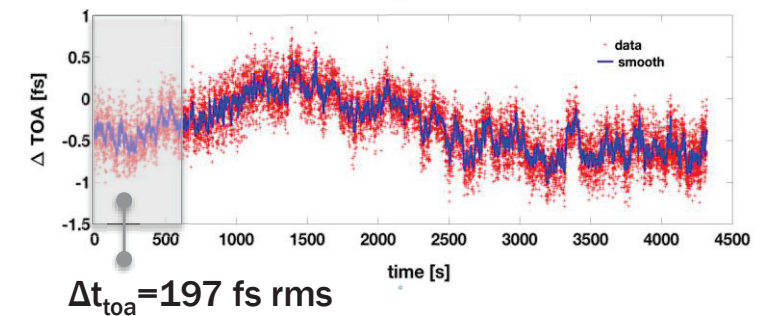


Collab: Stanford MIPS



Optical gating of CdTe2 crystal

Collab: L. Doolittle, G. Huang

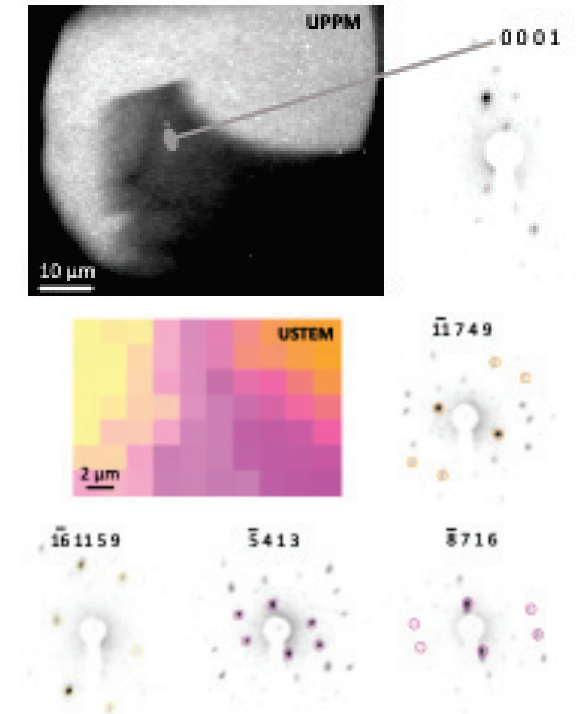
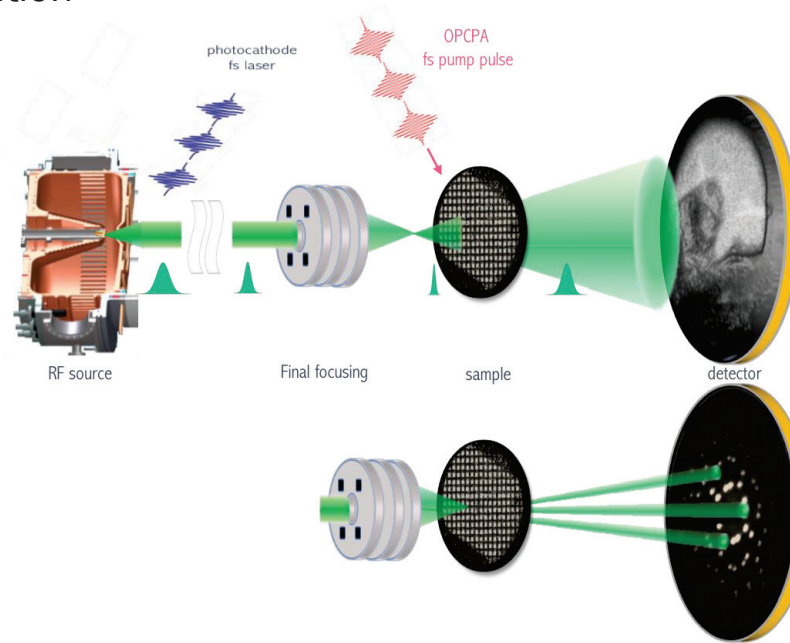
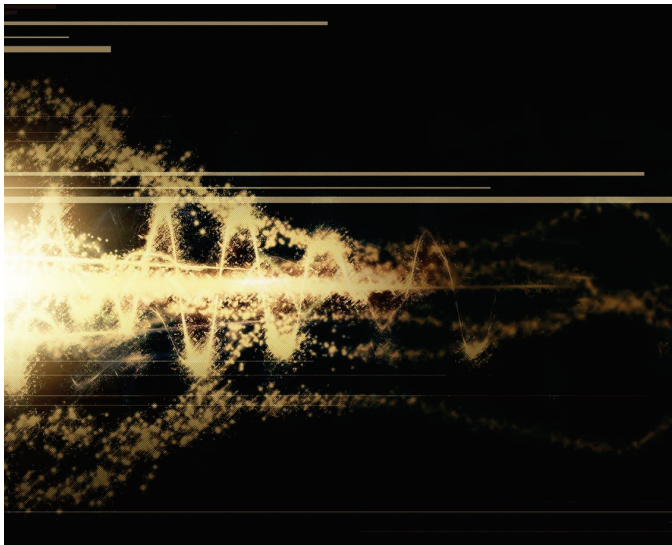


Conclusions

- HiRES is a high average brightness beamline, highly flexible for solid/gas/UED/UEM applications
- Electron probes with sub-nm emittance, sub-um beam size realized
- Knife-edge scan technique developed to precisely reconstruct the full 4D beam matrix of sub-nm emittance beams
- Multiple applications demonstrated for the study of structure dynamics in sub-micron crystal grains and functional nano-structures
- Further R&D to push the temporal/spatial resolution

*“Ultrafast relativistic electron Nanoprobes”,
F. Ji et al., [Commun. Phys. 2, 54 \(2019\)](#)*

*“Knife-edge based measurement of the 4D
transverse phase space”, F. Ji et al., [Phys. Rev.
ST Accel. Beams 22, 082801 \(2019\)](#)*



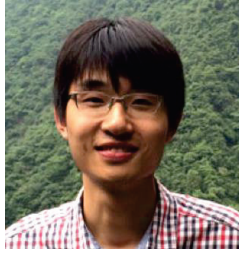
“Probing matter with relativistic electron beams”, [Nat. Revs. Phys. 1, 363 \(2019\)](#)

Team and acknowledgement

HIRES



Daniele Filippetto



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NanoUED



Pietro Musumeci
(UCLA)



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(NCEM/LBNL)



Dan Durham
(UCB/LBNL)



Jorge Giner Navarro
(UCLA)

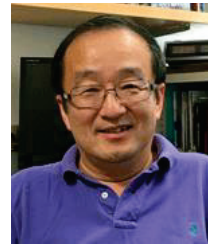
ATAP



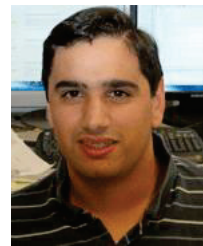
Larry Doolittle



Gang Huang



Derun Li



Diego Arbelaez

Quantum material



Robert Kaindl
(MSD/LBNL)



Khalid Siddique
(MSD/LBNL)

Gas Phase



Martin Centurion
(UNL)



Dan Slaughter
(CSD/LBNL)



Joshua
Williams(UNR)

- *LBNL-LDRD on NanoUED with NCEM*
- *LBNL-LDRD on quantum materials with MSD*
- *Samples received from UCB, and MF*
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